EQUIVARIANT HILBERT-NOETHERIAN FINITE GENERATION OVER WELL-STRUCTURED ORDERED DOMAINS AND ITS NECESSARY STRUCTURAL CONDITION

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Editorial scope. Only the named principal target counted. All original domain/effectiveness/reversibility conditions remain. Known public computable-presentation lemma, prior equality/ordered special cases, cited conjectures and direct linear-solvability application are uncounted context. Zeroness theorem is retained at its actual finite-testing scope.

Allowed standard prerequisites (editorial).

- Commutative algebra, Noetherian rings, well-quasi-orders, effective relational structures, register automata and reversible Petri nets. Complete original definitions and bibliography retained. Standard Hilbert/Buchberger and well-quasi-order facts are cited prerequisites; all new equivariant constructions and author proofs retained. Known public155220 computable-presentation lemma is supporting context, not a new task.

1. Introduction

A ring is *Noetherian* if each of its ideals is finitely-generated. According to Hilbert's Basis Theorem, the ring of polynomials over a finite set of variables, with coefficients from a given Noetherian ring, is Noetherian. In particular, every polynomial ideal can be represented by its finite basis, and hence can be given as input to algorithms. Given such a finite basis, Buchberger's algorithm computes a Gröbner basis that can be used to decide the membership problem for polynomial ideals: determine, whether a given polynomial belongs to the ideal generated by a given set of polynomials. Hilbert's Basis Theorem and Gröbner bases have important algorithmic applications, for instance to decide zeroness of polynomial automata (weighted automata with polynomial updates) [3], reachability of reversible Petri nets [24], or bisimulation equivalence checking [27]. Other applications of Buchberger's algorithm are overviewed in [8].

We generalise the above classical results to the ring of polynomials over an infinite set of variables. Clearly, Hilbert's Basis Theorem is not generalisable literally, since every polynomial belonging to a finitely-generated ideal may use at least one of variables used by the generators. In this paper we investigate the following setting:

- Variables are elements of a countable relational structure $\mathcal{A}$ (which we call *domain of variables*).
- The monoid of embeddings $\mathcal{A} \to \mathcal{A}$ acts on polynomials by renaming variables.
- We restrict to polynomial ideals which are invariant under renaming (such ideals we call *equivariant*). Accordingly, we adapt the definition of basis, enforcing equivariance of a generated ideal.

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Example 1. For illustration, consider *equality* domain of variables, namely the structure $\mathcal{A} = (A, =)$ consisting of a countable infinite set A and the equality relation. Embeddings are injective mappings $A \to A$. Consider the ring $\mathbb{R}[\mathcal{A}]$ of polynomials with real coefficients over variables A. The subset $\mathcal{I} \subseteq \mathbb{R}[\mathcal{A}]$ of polynomials whose coefficients sum up to 0, is an equivariant ideal. It is the least ideal containing all univariate polynomials $p_a(a) = a - 1$, where $a \in A$. For instance, the polynomial $p(a, b, c) = ab^2 - bc$ with variables $a, b, c \in A$ belongs to $\mathcal{I}$, and decomposes as:

$$ab^2 - bc \;=\; (b-1)(ab + a - c) + (a-1) - (c-1).$$

$\mathcal{I}$ is also the least *equivariant* ideal including $\{p_a\}$, for a fixed $a \in A$, hence finitely-generated according to our adapted definition. ◁

Contribution. We investigate equivariant ideals of polynomials over domains of variables which are *totally ordered*, namely admit a total order as one of its relations, and *well structured*, namely admit a naturally defined *well quasi order* (as defined in Section 2). A prototypical example of such a domain is rational order $\mathcal{Q} = (Q, \leq_{\mathcal{Q}})$. For such domains $\mathcal{A}$ we prove, in Section 3, the following generalisation of Hilbert's Basis Theorem: for every Noetherian ring $\mathbb{K}$, every equivariant ideal in the polynomial ring $\mathbb{K}[\mathcal{A}]$ has a finite basis. We transfer this property to lexicographic products of totally ordered, well structured domains, and to reducts thereof, including the equality domain. We also show that well structure is necessary for the generalisation of Hilbert's Basis Theorem to hold.

For further investigations we restrict to domains of variables which are *well ordered*, well structured, and *computable* (as formalized in Section 5). We develop a generalisation of Buchberger's algorithm to compute a Gröbner basis of an equivariant ideal, given as a finite set of generators. As a corollary, in Section 7 we obtain decidability of the membership problem for equivariant ideals. Similarly as our generalisation of Hilbert's Basis Theorem, we transfer the decidability result to a wide range of other domains, including equality domain, rational order, and lexicographic products thereof.

Our results are of fundamental nature, with various potential applications. We sketch upon some of them. In Section 4 we apply our generalisation of Hilbert's Basis Theorem to prove that orbit-finitely generated vector spaces have finite length, i.e., admit no infinite increasing chains of equivariant subspaces. This generalises a crucial result of [6] (however, without explicit upper bound). As a consequence, we derive decidability of the zeroness problem for weighted register automata, again generalising a result of [6]. We also apply, in Section 8, the ideal membership problem to decide reachability in reversible Petri nets with data, for a wide range of data domains. This includes rational order domain, where the problem is known to be undecidable for general non-reversible Petri nets [23]. Finally, in Section 9, we apply the ideal membership problem to solving orbit-finite systems of linear equations. This generalises a core result of [15] to a wide range of domains (again, without complexity upper bound). All our proofs of generalisations of [6] and [15] are remarkably simple, compared to the original ones.

Structure of the paper. The paper has two parts: the first part concentrates on *existence* of finite bases, while the second one investigates *computability* thereof. In the first part, Sections 2–4, we set up preliminaries, prove our generalisation of Hilbert's Basis Theorem, and sketch upon its applications. Then, in Sections 5–9 we generalise Buchberger algorithm, use it for decidabillity of the ideal membership problem, and show applications thereof. Missing proofs are to be found in the full version of this paper [16].

Related research. Our motivations, together with the most fundamental concepts such as equivariance and orbit-finiteness, come from the theory of sets with atoms [5, 4]. The concept of well structured domain, crucial for development of this paper, turns out to be essentially the same as *preservation of well quasi order*, introduced in [22] and conjectured to characterise data domains rendering decidability of a range of decision problems for Petri nets. Relationship between the two settings calls for further investigation.

There have been numerous earlier contributions regarding existence and computability of finite bases of equivariant ideals. To the best of our knowledge, all of them considered (using our terminology) either equality or well order domains. First, [10] proves finite basis property for domain $\omega = (\mathbb{N}, \leq)$, then [11] and [14] extend to larger ordinals. These two papers also give sufficient condition for computability of Gröbner bases, which is shown to hold in ω in [18]. Similar results were independently obtained in [1, 2, 7]. In [19] the authors use the aforementioned results to prove the Independent Set Conjecture in algebraic statistics.

There are a number of recent results concerning finite generation of vector spaces: [6] shows that every equivariant subspace of orbit-finitely generated vector space is finitely generated (this follows from our results in Section 3, as argued in Section 4); furthermore, [15] shows that dual vector spaces are finitely generated.

2. Preliminaries

A relational structure $\mathcal{A} = (A, r_1, \ldots, r_n)$ is a set A together with finitely many relations $r_1 \subseteq A^{s_1}, \ldots, r_n \subseteq A^{s_n}$ of finite arities $s_1, \ldots, s_n \in \mathbb{N}$, respectively. In the sequel we restrict to countable structures only. Every relation $r \subseteq A^m$ which is expressible as a Boolean combination of relations $r_1, \ldots, r_n$, we call *definable*. We always assume that equality $=$ is definable. An *induced substructure* of $\mathcal{A}$ is the restriction of $\mathcal{A}$ to some subset of A.

An *embedding* of $\mathcal{A}$ is a map $\iota : A \to A$ that preserves and reflects all relations, which means that for every relation r_i and $a_1, \ldots, a_{s_i} \in A$, the following equivalence holds:

$$r_i(a_1, \ldots, a_{s_i}) \iff r_i(\iota(a_1), \ldots, \iota(a_{s_i})).$$

In other words, an embedding ι is an isomorphism of $\mathcal{A}$ and its substructure induced by $\iota(A)$. Clearly, embeddings preserve and reflect all definable relations, and hence each such relation can be considered w.l.o.g. to be among $r_1, \ldots, r_n$. The set of all embeddings of $\mathcal{A}$ we denote as $\textsc{Emb}_{\mathcal{A}}$. We say that $\mathcal{A}$ is *totally ordered* (resp. *well ordered*) if a total order (resp. a well order) is definable in $\mathcal{A}$.

Speaking intuitively, a *reduct* of $\mathcal{A}$ is a structure obtained by removing some relations from $\mathcal{A}$. Formally, a structure $\mathcal{A}' = (A, r_1, \ldots, r_n)$ is a reduct of $\mathcal{A} = (A, \ldots)$, with the same carrier set A, if all relations $r_1, \ldots, r_n$ are definable in $\mathcal{A}$.

Example 2. For a countable infinite set A, embeddings of the equality structure $\mathcal{A} = (A, =)$, with equality as the only relation, are all injective maps $A \to A$. Embeddings of the rational order $\mathcal{Q} = (Q, \leq_{\mathcal{Q}})$ are all strictly monotonic maps $\iota : Q \to Q$, i.e. functions satisfying

$$q <_{\mathcal{Q}} q' \implies \iota(q) <_{\mathcal{Q}} \iota(q').$$

Note that equality is definable using $\leq$, namely $q = q' \iff q \leq q' \wedge q' \leq q$. The equality structure is thus a reduct of $\mathcal{Q}$.

Every ordinal α can be seen as a (well ordered) relational structure, with just one binary relation – the ordinal order. Again, its embeddings are strictly monotonic maps $\alpha \to \alpha$. A special case is $\omega = (\mathbb{N}, \leq) = \{0 < 1 < \ldots\}$; or $\langle d \rangle = \{0 < 1 < \ldots < d{-}1\}$, for $d \in \mathbb{N}$, where the only embedding is the identity. In the sequel we identify an ordinal with the corresponding relational structure. ◁

Noetherianity. Recall that an *ideal* in a commutative ring $\mathbb{K}$ is any nonempty subset $\mathcal{I} \subseteq \mathbb{K}$ closed under addition and multiplication by any element of $\mathbb{K}$: $a, a' \in \mathcal{I} \implies a+a' \in \mathcal{I}$, and $a \in \mathcal{I}, b \in \mathbb{K} \implies a \cdot b \in \mathcal{I}$. A basis of $\mathcal{I}$ is any subset $\mathcal{B} \subseteq \mathcal{I}$ such that every $a \in \mathcal{I}$ is obtainable from $\mathcal{B}$ by addition and multiplication by arbitrary elements of $\mathbb{K}$; equivalently, $a = a_1 \cdot b_1 + \ldots + a_n \cdot b_n$ for some $b_1, \ldots, b_n \in \mathcal{B}$ and $a_1, \ldots, a_n \in \mathbb{K}$. A commutative ring is *Noetherian* if each of its ideals has a finite basis. In particular, each field is a Noetherian commutative ring.

Polynomials. Let $\mathcal{A} = (A, r_1, \ldots, r_n)$ be a relational structure, which we call a *domain of variables*, or simply a *domain*. We define, in the expected way, the commutative multiplicative monoid of monomials over variables from A, denoted $\textsc{Mon}[\mathcal{A}]$. A *monomial* is a formal expression of the form

$$f(a_1 \ldots a_m) = a_1^{n_1} \cdot \ \ldots \ \cdot a_m^{n_m},$$

where $m \geq 0$, $\textsc{vars}(f) = \{a_1, \ldots, a_m\} \subseteq A$ are distinct variables, and $n_i = \textsc{deg}_f\, a_i > 0$ is the *degree* of variable a_i in f. For $a \notin \textsc{vars}(f)$ we set $\textsc{deg}_f\, a = 0$. The trivial unit monomial, for $m = 0$, we denote as $\mathbf{1}$. Also, given a commutative ring $\mathbb{K}$, we define the commutative ring of polynomials with coefficients from $\mathbb{K}$ and variables from A, denoted $\mathbb{K}[\mathcal{A}]$. A *term* is a formal expression of the form $r \cdot f$, for $r \in \mathbb{K}$ and $f \in \textsc{Mon}[\mathcal{A}]$, and a *polynomial* is a finite sum of terms, i.e., a formal expression of the form

$$f \ = \ r_1 \cdot f_1 + \ldots + r_n \cdot f_n,$$

$f_1, \ldots, f_n \in \textsc{Mon}[\mathcal{A}]$ are distinct monomials and $r_1, \ldots, r_n \in \mathbb{K}$ are their *coefficients*. The zero polynomial, for $n = 0$, we denote by $\mathbf{0}$. We set $\textsc{vars}(f) = \textsc{vars}(f_1) \cup \ldots \cup \textsc{vars}(f_n)$.

Equivariant Noetherianity. Embeddings act on polynomials by renaming variables: an embedding $\iota \in \textsc{Emb}_A$ maps a polynomial f to the polynomial $\iota(f)$ obtained by renaming all variables in f according to ι, i.e., by replacing each variable a by $\iota(a)$[1]. An ideal $\mathcal{I} \subseteq \mathbb{K}[\mathcal{A}]$ is called *equivariant* if it is additionally closed under the action of $\textsc{Emb}_{\mathcal{A}}$: whenever $f \in \mathcal{I}$ and $\iota \in \textsc{Emb}_{\mathcal{A}}$ then $\iota(f) \in \mathcal{I}$. We adapt the notion of basis by taking additionally into account the action of $\textsc{Emb}_{\mathcal{A}}$. By a *basis* of an equivariant ideal $\mathcal{I}$ we mean a subset $\mathcal{B} \subseteq \mathcal{I}$ such that each element of $\mathcal{I}$ is obtainable from $\mathcal{B}$ by addition, multiplication by arbitrary polynomials, and action of $\textsc{Emb}_{\mathcal{A}}$. This is equivalent to saying that

$$\textsc{Emb}_{\mathcal{A}}(\mathcal{B}) \ \stackrel{\text{def}}{=} \ \{\iota(g) \,|\, \iota \in \textsc{Emb}_{\mathcal{A}},\ g \in \mathcal{B}\} \tag{2.1}$$

is a basis of $\mathcal{I}$ in classical sense, or to presentability of every element $f \in \mathcal{I}$ as

$$f \ = \ \sum_{i=1}^{n} h_i \cdot \iota_i(g_i),$$

for $g_1 \ldots g_n \in \mathcal{B}$ and $h_1 \ldots h_n \in \mathbb{K}[\mathcal{A}]$ and $\iota_1 \ldots \iota_n \in \textsc{Emb}_{\mathcal{A}}$ (since the action of $\textsc{Emb}_{\mathcal{A}}$ commutes with addition and multiplication of polynomials). When $\mathcal{B}$ is a basis of $\mathcal{I}$, we also say that $\mathcal{I}$ is the equivariant ideal *generated* by $\mathcal{B}$, and denote it by $\mathcal{I} = \langle\mathcal{B}\rangle$.

Example 3. Consider the rational order $\mathcal{Q} = (Q, \leq_{\mathcal{Q}})$ as a domain of variables, and a polynomial

$$f_0(a_0, b_0, c_0) \ = \ b_0^2 - a_0 c_0$$

for some arbitrarily chosen, fixed variables $a_0 < b_0 < c_0$ from Q. For every other variables $a < b < c$, the polynomial $ac^2 - b^3$ belongs to the equivariant ideal generated by $\{f_0\}$, because

$$ac^2 - b^3 \ = \ a \cdot (c^2 - bd) \ - \ b \cdot (b^2 - ad),$$

[1]Clearly, f and $\iota(f)$ define the same functions $\mathbb{K}^n \to \mathbb{K}$, where $n = |\textsc{vars}(f)|$.

for arbitrary variable $d > c$. ◁

The ring $\mathbb{K}[\mathcal{A}]$ we call *equivariantly Noetherian* if each equivariant ideal has a finite basis. In Section 3 we demonstrate that a wide range of domains $\mathcal{A}$ satisfy the following generalisation of Hilbert's Basis Theorem:

Property 4. *For every Noetherian commutative ring $\mathbb{K}$, the polynomial ring $\mathbb{K}[\mathcal{A}]$ is equivariantly Noetherian.*

Orbit-finite bases. Equivalently, $\mathbb{K}[\mathcal{A}]$ is equivariantly Noetherian if each equivariant ideal has a basis, in classical sense, which is *orbit-finite*, i.e., finite up to embeddings. Formally, orbit-finite sets are set of the form (2.1) where $\mathcal{B}$ is finite. A finite set $\mathcal{B}$ is called a *presentation* of an orbit-finite set (2.1). The concept of orbit-finiteness comes from the theory of sets with atoms [4, 5], where it refers to sets which are infinite but finite up to automorphisms of an underlying structure of atoms. We slightly extend this setting here, and consider embeddings instead of just automorphisms. In case of equality and rational order domains (and any *homogeneous* structure), the action of embeddings is essentially the same as the action of automorphisms, as every automorphism of finite induced substructures extends to an automorphism of the whole domain.

Ideal membership. For $\mathcal{A}$ satisfying Property 4, every equivariant ideal in $\mathbb{K}[\mathcal{A}]$ can be represented by its finite basis, and thus can be input to algorithms. In Section 7, assuming $\mathbb{K}$ to be a fixed computable field (e.g. rational[2] field $\mathbb{Q}$), we study decidability of the following decision problem of membership of a given polynomial in a given equivariant ideal, for different domains $\mathcal{A}$:

IDEAL-MEMB IN $\mathbb{K}[\mathcal{A}]$**::**
Input: : a polynomial $f \in \mathbb{K}[\mathcal{A}]$ and an equivariant ideal
: $\mathcal{I} \subseteq \mathbb{K}[\mathcal{A}]$.
Question: : $f \in \mathcal{I}$?

Well structured structures. A quasi order consists of a set Y and a reflexive and transitive relation $\preceq \subseteq Y \times Y$. $(Y, \preceq)$ is a *well quasi order* (WQO) if every infinite sequence $y_1, y_2, \ldots$ of elements of Y contains a dominating pair, i.e., two elements y_i and y_j, for $i < j$, such that $y_i \preceq y_j$. Clearly, every well order is a WQO.

A subset $X \subseteq Y$ is *upward-closed* if $x \in X$ and $x \preceq x'$ implies $x' \in X$. A *basis* of an upward-closed set X (not to be confused with a basis of an ideal) is any subset $B \subseteq X$ such that for every $x \in X$ there is $b \in B$ with $b \preceq x$. Every WQO is well-founded and admits no infinite antichains [13, Proposition 12.1.1], which implies:

Lemma 5. *Every upward-closed subset of a* WQO *has a finite basis.*

Given a relational structure $\mathcal{A}$ and a quasi order $(Y, \preceq)$, by $\mathrm{AGE}(\mathcal{A})$ we denote the set of all its finite induced substructures. We do not distinguish between a finite subset $B \subseteq_{\mathrm{fin}} A$ and the substructure $\mathcal{B}$ of $\mathcal{A}$ induced by B, i.e., we equate $\mathrm{AGE}(\mathcal{A})$ with $\mathcal{P}_{\mathrm{fin}}(\mathcal{A})$. By $\mathrm{AGE}(\mathcal{A}, Y)$ we denote the set of all finite induced substructures labeled by Y, i.e., functions $\ell : B \to Y$ where $B \in \mathrm{AGE}(\mathcal{A})$. The quasi order $(Y, \preceq)$ *determines* a natural quasi order $\rightsquigarrow$ on $\mathrm{AGE}(\mathcal{A}, Y)$, where $(\ell : B \to Y) \rightsquigarrow (\ell' : B' \to Y)$ if there is an embedding $\iota \in \mathrm{EMB}_{\mathcal{A}}$ such that $\iota(B) \subseteq B'$ and $\ell(b) \preceq \ell'(\iota(b))$ for all $b \in B$. In most cases we will focus on $(Y, \preceq) = (\mathbb{N}, \leq)$.

Definition 6. $\mathcal{A}$ is *ω-well structured* if $\omega = (\mathbb{N}, \leq)$ determines a WQO on $\mathrm{AGE}(\mathcal{A}, \mathbb{N})$. Moreover, $\mathcal{A}$ is *well structured* if every WQO $(Y, \preceq)$ determines a WQO on $\mathrm{AGE}(\mathcal{A}, Y)$.[3] ◁

[2]The set of rationals appears in the sequel in two roles: as a totally ordered set $\mathcal{Q}$, and as a field $\mathbb{Q}$. We use different fonts to distinguish these two roles.

[3]It is possible that the two conditions are equivalent. See e.g. [21] for related results.

For instance, the rational order $\mathcal{Q} = (Q, \leq_{\mathcal{Q}})$ is well structured. Indeed, finite induced substructures of $\mathcal{Q}$ are finite total orders, and $\textsc{Age}(\mathcal{Q}, Y)$ is a WQO by Higman's Lemma [17]. The equality structure is also well structured, as a reduct of $\mathcal{Q}$. *Universal tree* arising as Fraïssé limit of finite trees [4, Section 7], similarly as $\mathcal{Q}$ arises as Fraïssé limit of finite total orders, is well structured. By Nash-Williams's Theorem [25, Theorem 2] (cf. also [17] in the restricted case of ω) ordinals are well structured too:

Lemma 7. *Every ordinal (seen as a relational structure) is well structured.*

Proof. Let α be a fixed ordinal and $(Y, \preceq)$ a fixed WQO. We add to Y the least element $*$, namely put $* \prec y$ for all $y \in Y$, and observe that $(Y \cup \{*\}, \preceq)$ is still a WQO. According to Theorem 2 in [25], the WQO $(Y \cup \{*\}, \preceq)$ determines a WQO on the set $F(Y)$ of image-finite functions $\ell : \alpha \to Y \cup \{*\}$. Elements of $\textsc{Age}(\alpha, Y)$ may be identified with those functions $\ell : \alpha \to Y$ which map almost all elements to 0, a subset $F_0(Y) \subseteq F(Y)$. Furthermore, the quasi orders determined by $(Y \cup \{*\}, \preceq)$ on $\textsc{Age}(\alpha, Y)$ and $F_0(Y)$ agree. Therefore $(Y \cup \{*\}, \preceq)$ determines a WQO on $\textsc{Age}(\alpha, Y)$, as required. ∎

Lexicographic product of structures. We will build larger domains from smaller ones, using *lexicographic product* defined below. The operation, in contrast to Cartesian product (as illustrated in Example 11), preserves well structure.

Given two relational structures $\mathcal{A} = (A, \ldots)$ and $\mathcal{A}' = (A', \ldots)$, their lexicographic product $\mathcal{A} \otimes \mathcal{A}'$ is defined as follows. The carrier set of $\mathcal{A} \otimes \mathcal{A}'$ is the Cartesian product $A \times A'$, and the set of relation symbols is the disjoint union of sets of relation symbols of $\mathcal{A}$ and $\mathcal{A}'$. For each k-ary relation r of $\mathcal{A}$, the relation r in $\mathcal{A} \otimes \mathcal{A}'$ relates those k-tuples

$$((a_1, a'_1), \ldots, (a_k, a'_k)) \tag{2.2}$$

where $r(a_1, \ldots, a_k)$ in $\mathcal{A}$ (the second coordinate comming from $\mathcal{A}'$ is ignored). Furthermore, for each k-ary relation r' of $\mathcal{A}'$, the relation r' in $\mathcal{A} \otimes \mathcal{A}'$ relates those tuples (2.2) where $a_1 = \ldots = a_k$ and $r'(a'_1, \ldots, a'_k)$ in $\mathcal{A}'$ (the first coordinate comming from $\mathcal{A}$ is required to be equal). The product is associative.

Lemma 8. *Embeddings of $\mathcal{A} \otimes \mathcal{A}'$ are exactly maps of the form*

$$(a, a') \longmapsto (\iota(a), \iota'_a(a')), \tag{2.3}$$

determined by some $\iota \in \textsc{Emb}_{\mathcal{A}}$ and a family $\{\iota'_a \in \textsc{Emb}_{\mathcal{A}'} \mid a \in A\}$.

Proof. In one direction, every mapping $A \times A' \to A \times A'$ of the form (2.3) preserves and reflects all relations of $\mathcal{A} \otimes \mathcal{A}'$, since each relation of $\mathcal{A}$ is preserved and reflected by $\iota \in \textsc{Emb}_{\mathcal{A}}$, and each relation of $\mathcal{A}'$ is preserved and reflected by every ι'_a, for $a \in A$.

In the reverse direction, consider any embedding $\kappa \in \textsc{Emb}_{A \otimes A'}$. As equality is definable in $\mathcal{A}$, the mapping κ preserves and reflects equality of the first coordinate, which determines an injection $\iota : A \to A$ such that the first component of $\kappa(a, a')$ equals $\iota(a)$. As κ preserves and reflects all relations of $\mathcal{A}$, the injection is an embedding, $\iota \in \textsc{Emb}_{\mathcal{A}}$. Furthermore, for each $a \in A$ the mapping κ restricted to pairs $(a, _)$, preserves and reflects all relations of $\mathcal{A}'$ on the second coordinate. Therefore, for each $a \in \mathcal{A}$ there is some $\iota'_a \in \textsc{Emb}_{\mathcal{A}'}$ such that κ is equal to (2.3). ∎

Lemma 9. *Lexicographic product of two totally ordered (well ordered, well structured, resp.) relational structures is totally ordered (well ordered, well structured, resp.).*

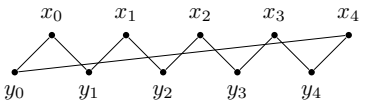


FIGURE 1. $C_5 \in \text{AGE}(\mathcal{Q} \times \mathcal{Q})$. Its elements are depicted by edges.

Proof. Consider two relational structures $\mathcal{A} = (A, \ldots)$ and $\mathcal{A}' = (A', \ldots)$. Let $\preceq$ and $\preceq'$ be definable total orders on $\mathcal{A}$ and $\mathcal{A}'$, respectively. The lexicographic product of $\preceq$ and $\preceq'$ is definable in $\mathcal{A} \otimes \mathcal{A}'$, and is well when $\preceq$ and $\preceq'$ are well, and therefore if both $\mathcal{A}$ and $\mathcal{A}'$ are totally ordered (resp. well ordered), so is $\mathcal{A} \otimes \mathcal{A}'$.

It thus remains to show that $\mathcal{A} \otimes \mathcal{A}'$ is well structured. This follows immediately, once we prove:

Claim 10. *Let $(Y, \prec)$ is a* WQO. *Then* $\text{AGE}(\mathcal{A} \otimes \mathcal{A}', Y)$ *is isomorphic, as a quasi order, to* $\text{AGE}(\mathcal{A}, \text{AGE}(\mathcal{A}', Y))$.

For proving the claim, consider any $B \in \text{AGE}(\mathcal{A} \otimes \mathcal{A}')$ and its labelling $(\ell : B \to Y) \in \text{AGE}(\mathcal{A} \otimes \mathcal{A}', Y)$. Let $\overline{B} \in \text{AGE}(\mathcal{A})$ be its projection on the first coordinate. Furthermore, for $a \in \overline{B}$ let $B_a \stackrel{\text{def}}{=} \{a' \mid (a, a') \in B\} \in \text{AGE}(\mathcal{A}')$, and let $\ell_a : B_a \to Y$ be the corresponding restriction of ℓ, namely $\ell_a(a') = \ell(a, a')$. This defines a function

$$\overline{\ell} : a \mapsto \ell_a, \qquad \text{for } a \in \overline{B},$$

i.e., $(\overline{\ell} : \overline{B} \to \text{AGE}(\mathcal{A}', Y)) \in \text{AGE}(\mathcal{A}, \text{AGE}(\mathcal{A}', Y))$. We have thus defined a mapping $\ell \mapsto \overline{\ell}$. One easily defines its inverse, and hence it is a bijection between $\text{AGE}(\mathcal{A} \otimes \mathcal{A}', Y)$ and $\text{AGE}(\mathcal{A}, \text{AGE}(\mathcal{A}', Y))$.

Given two induced substructures $B_i \in \text{AGE}(\mathcal{A} \otimes \mathcal{A}')$ and its labellings $(\ell_i : B_i \to Y) \in \text{AGE}(\mathcal{A} \otimes \mathcal{A}', Y)$, for $i = 1, 2$, using Lemma 8 we notice equivalence of the following two conditions:

- $\ell_1 \rightsquigarrow \ell_2$ are related by the quasi order determined by $(Y, \preceq)$ on $\text{AGE}(\mathcal{A} \otimes \mathcal{A}', Y)$;
- there is some $\iota \in \text{EMB}_{\mathcal{A}}$ such that for every $a \in \overline{B}$, $(\ell_1)_a \rightsquigarrow (\ell_2)_{\iota(a)}$ are related by the quasi order determined by $(Y, \preceq)$ on $\text{AGE}(\mathcal{A}', Y)$.

The latter condition says exactly that $\overline{\ell}_1 \rightsquigarrow \overline{\ell}_2$ are related by the quasi order determined by $(Y, \preceq)$ on $\text{AGE}(\mathcal{A}, \text{AGE}(\mathcal{A}', Y))$. The claim is thus shown. ■

Example 11. Since the rational order $\mathcal{Q}$ is well structured, the lexicographic product $\mathcal{Q} \otimes \mathcal{Q}$ is also so, while the Cartesian product $\mathcal{Q} \times \mathcal{Q}$ is not. We claim that the quasi order $\rightsquigarrow$ determined by an arbitrary WQO on $\text{AGE}(\mathcal{Q} \times \mathcal{Q})$ (the choice of WQO is irrelevant) is not a WQO. Indeed, consider the family of 'cycle' subsets $C = (C_n \in \text{AGE}(\mathcal{Q} \times \mathcal{Q}))_{n \in \mathbb{N}}$, where (cf. Figure 1)

$$C_n = \{(x_i, y_i), (x_i, y_{i+1 \bmod n}) \mid i \in \{0 \ldots n-1\}\}$$

for arbitrary rationals $x_0 < \ldots < x_{n-1}$ and $y_0 < \ldots < y_{n-1}$. As embeddings of $\mathcal{Q} \times \mathcal{Q}$ are induced by pairs of embeddings of $\mathcal{Q}$,

$$\text{EMB}_{\mathcal{Q} \times \mathcal{Q}} = \text{EMB}_{\mathcal{Q}} \times \text{EMB}_{\mathcal{Q}},$$

a smaller cycle can not embed into a larger one. Therefore $C_n \rightsquigarrow C_m$ implies $n = m$, i.e., C is an antichain, and hence $\rightsquigarrow$ is not a WQO. ◁

3. Equivariant Noetherianity

We prove that every totally ordered and ω-well structured domain of variables satisfies Property 4. Moreover, almost conversely, Property 4 implies that domain of variables is ω-well structured.

Theorem 12. *If the polynomial ring $\mathbb{K}[\mathcal{A}]$ is equivariantly Noetherian for some commutative ring $\mathbb{K}$ then $\mathcal{A}$ is ω-well structured.*

Theorem 13. *For every ω-well structured and totally ordered domain of variables $\mathcal{A}$ and a Noetherian commutative ring $\mathbb{K}$, the polynomial ring $\mathbb{K}[\mathcal{A}]$ is equivariantly Noetherian.*

Proof. Suppose, towards contradiction, that an equivariant ideal $\mathcal{I} \subseteq \mathbb{K}[\mathcal{A}]$ is not finitely generated. Then there exists an infinite sequence $f_0, f_1, f_2, \dots$ of polynomials in $\mathcal{I}$ such that $f_n \in \mathcal{I} \setminus \langle\{f_0, \dots, f_{n-1}\}\rangle$ for all $n \in \mathbb{N}$. Since $\preceq_{\text{lex}}$ is well-founded we can furthermore assume that $\textsc{cp}(f_n)$ is $\preceq_{\text{lex}}$-minimal for all n, i.e. there is no $g \in \mathcal{I} \setminus \langle\{f_1, \dots, f_{n-1}\}\rangle$ such that $\textsc{cp}(g) \prec_{\text{lex}} \textsc{cp}(f_n)$. As $\sqsubseteq$ is a well partial order, the sequence $(\textsc{cp}(f_n))_{n\in\mathbb{N}}$ contains an infinite non-decreasing subsequence:

$$\textsc{cp}(f_{n_1}) \;\sqsubseteq\; \textsc{cp}(f_{n_2}) \;\sqsubseteq\; \dots$$

Let $\mathcal{J}_k \subseteq \mathbb{K}$ be the ideal generated by $\{\textsc{lc}(f_{n_1}), \dots, \textsc{lc}(f_{n_k})\}$. We have thus an increasing chain of ideals in $\mathbb{K}$,

$$\mathcal{J}_1 \;\subseteq\; \mathcal{J}_2 \;\subseteq\; \dots,$$

which necessarily stabilises as $\mathbb{K}$ is Noetherian. Let $K \in \mathbb{N}$ be such that $\mathcal{J}_n = \mathcal{J}_K$ for all $n \geq K$. As $\mathcal{J}_{K+1} = \mathcal{J}_K$, there are $r_1, \dots, r_K \in \mathbb{K}$ such that

$$\textsc{lc}(f_{n_{K+1}}) = \sum_{i=1}^{K} r_i \cdot \textsc{lc}(f_{n_i}). \tag{3.1}$$

Moreover, since $\textsc{cp}(f_{n_i}) \sqsubseteq \textsc{cp}(f_{n_{K+1}})$ for $i = 1 \dots K$, there are some monomials $g_1, \dots, g_K \in \textsc{Mon}[\mathcal{A}]$ and embeddings $\iota_1, \dots, \iota_K \in \textsc{Emb}_{\mathcal{A}}$ such that

$$g_i \cdot \iota_i(\textsc{lm}(f_{n_i})) = \textsc{lm}(f_{n_{K+1}})$$
$$\iota_i(\textsc{vars}(f_{n_i})) \subseteq \textsc{vars}(f_{n_{i+1}})$$

for $i = 1 \dots K$. Using the fact that $\textsc{lm}(_)$ and $\textsc{vars}(_)$ both commute with the action of embeddings, and Lemma 19, the left-hand side can be rewritten to:

$$\textsc{lm}(g_i \cdot \iota_i(f_{n_i})) = \textsc{lm}(f_{n_{K+1}}) \tag{3.2}$$
$$\textsc{vars}(\iota_i(f_{n_i})) \subseteq \textsc{vars}(f_{n_{i+1}}) \tag{3.3}$$

Since $\textsc{lt}(f_{n_i}) = \textsc{lc}(f_{n_i}) \cdot \textsc{lm}(f_{n_i})$, by the equalities (3.1) and (3.2) we get:

$$\textsc{lt}(f_{n_{K+1}}) \;=\; \sum_{i=1}^{K} r_i \cdot \textsc{lt}(g_i \cdot \iota_i(f_{n_i})).$$

Using the equality $r \cdot \textsc{lt}(f) = \textsc{lt}(r \cdot f)$, we rewrite further to:

$$\textsc{lt}(f_{n_{K+1}}) \;=\; \sum_{i=1}^{K} \textsc{lt}(r_i \cdot g_i \cdot \iota_i(f_{n_i})).$$

The left-hand side of the above equality contains just one term, therefore $\textsc{lm}(f_{n_{K+1}}) = \textsc{lm}(r_i \cdot g_i \cdot \iota_i(f_{n_i}))$ for some $i \in \{1 \dots K\}$, and leading monomials different that $\textsc{lm}(f_{n_{K+1}})$ cancel out in the right-hand side sum. Dropping summands with leading monomials different that $\textsc{lm}(f_{n_{K+1}})$ yields:

$$\textsc{lt}(f_{n_{K+1}}) \;=\; \textsc{lt}\Big(\sum_{i=1}^{K} r_i \cdot g_i \cdot \iota_i(f_{n_i})\Big). \tag{3.4}$$

Let f' be the polynomial appearing on the right-hand side:

$$f' \;:=\; \sum_{i=1}^{K} r_i \cdot g_i \cdot \iota_i(f_{n_i}).$$

As $f' \in \mathcal{I}_{n_K}$ and $f_{n_{K+1}} \in \mathcal{I}_{n_{K+1}} \setminus \mathcal{I}_{n_K}$, the difference of polynomials $(f_{n_{K+1}} - f') \in \mathcal{I}_{n_{K+1}} \setminus \mathcal{I}_{n_K}$. All the leading terms appearing on both sides of (3.4) cancel out in $f_{n_{K+1}} - f'$. We thus deduce

$$\textsc{lm}(f_{n_{K+1}} - f') \prec_{\text{lex}} \textsc{lm}(f_{n_{K+1}}).$$

Moreover, by (3.2) and (3.3) we deduce

$$\textsc{vars}(f_{n_{K+1}} - f') \subseteq \textsc{vars}(f_{n_{K+1}}) \ .$$

Combining the last two statements we obtain

$$\textsc{cp}(f_{n_{K+1}} - f') \prec_{\text{lex}} \textsc{cp}(f_{n_{K+1}}),$$

a contradiction with $\preceq_{\text{lex}}$-minimality of $\textsc{cp}(f_{n_{K+1}})$, which completes the proof. ■

Lemma 14. *If $\mathbb{K}[\mathcal{A}]$ is equivariantly Noetherian and $\mathcal{B}$ is a reduct of $\mathcal{A}$ then $\mathbb{K}[\mathcal{B}]$ is also equivariantly Noetherian.*

Proof. Since $\textsc{Emb}_{\mathcal{A}} \subseteq \textsc{Emb}_{\mathcal{B}}$, every equivariant ideal $\mathcal{I}$ in $\mathbb{K}[\mathcal{B}]$ is automatically an equivariant ideal in $\mathbb{K}[\mathcal{A}]$, and therefore $\mathcal{I}$ is finitely generated in there. Furthermore, again due to $\textsc{Emb}_{\mathcal{A}} \subseteq \textsc{Emb}_{\mathcal{B}}$, a finite basis of $\mathcal{I}$ in $\mathbb{K}[\mathcal{A}]$ is also its basis in $\mathbb{K}[\mathcal{B}]$. ■

Remark 15. Theorem 13 applies in particular to: every ordinal (by Lemma 7); rational order $\mathcal{Q}$; equality domain (by Lemma 14); lexicographic products thereof (by Lemma 9). All subsequent results in this paper apply to all the above-listed domains too.

According to Pouzet's conjecture [26], necessary and sufficient conditions of Theorems 12 and 13 coincide, namely every ω-well structured structure may be equipped with a total order while preserving ω-well structure:

Conjecture 16 ([26])**.** *Every ω-well structured relational structure $\mathcal{A}$ is a reduct of an ω-well structured, totally ordered structure.*

The conjecture, together with Theorems 12 and 13, and Lemma 14, would imply that Property 4 is satisfied exactly by reducts of ω-well structured and totally ordered domains of variables.

Quasi orders on monomials. We need some definitions before proving Theorems 12 and 13. In the sequel let $\mathbb{K}$ be a fixed Noetherian commutative ring, and let $\mathcal{A} = (A, \preceq, \ldots)$ be a fixed domain of variables, totally ordered by $\preceq$. We define three quasi orders on monomials, related by the following refinements:

$$f \preceq_{\text{lex}} f' \impliedby f \mid f' \implies f \sqsubseteq f'.$$

The two left ones are standard orders. Given two monomials f and f', *division order* is defined by $f \mid f'$ if $f \cdot g = f'$ for some $g \in \textsc{Mon}[\mathcal{A}]$. In order to define *lexicographic order*, we consider the $\preceq$-largest variable $a \in A$ such that $\textsc{deg}_f a \neq \textsc{deg}_{f'} a$, and set $f \prec_{\text{lex}} f'$ if $\textsc{deg}_f a < \textsc{deg}_{f'} a$. Lexicographic order is a *term order*:

Lemma 17. *For every monomials $f, f', g \in \textsc{Mon}[\mathcal{A}]$,*

$$f \mid f' \implies f \preceq_{lex} f' \implies f \cdot g \preceq_{lex} f' \cdot g.$$

The quasi order $\sqsubseteq$ is defined as relaxation of division order modulo renamings of variables:

$$f \sqsubseteq f' \iff \exists \iota \in \textsc{Emb}_{\mathcal{A}} : \iota(f) \mid f'. \tag{3.5}$$

We argue that (3.5) defines a quasi order, i.e., a transitive relation. Indeed, as action of embeddings preserves the division order, namely

$$f \mid f' \implies \kappa(f) \mid \kappa(f')$$

for every $\kappa \in \textsc{Emb}_{\mathcal{A}}$, we may deduce transitivity of $\sqsubseteq$:

$$\iota(f) \mid f' \ \wedge \ \kappa(f') \mid f'' \implies (\kappa \circ \iota)(f) \mid \kappa(f') \mid f''.$$

$\textsc{Mon}[\mathcal{A}]$ may be identified with $\textsc{Age}(\mathcal{A}, \mathbb{N} \setminus 0)$, and $\sqsubseteq$ with the quasi order determined by ω on $\textsc{Age}(\mathcal{A}, \mathbb{N} \setminus 0)$. Thus $\sqsubseteq$ is a WQO, if $\mathcal{A}$ is ω-well structured.

Definition 18. Given a non-zero polynomial $f \in \mathbb{K}[\mathcal{A}]$, $f \neq \mathbf{0}$, the *leading monomial* $\textsc{lm}(f) \in \textsc{Mon}[\mathcal{A}]$ of f is the $\preceq_{\text{lex}}$-largest monomial appearing in f; the *leading coefficient* $\textsc{lc}(f) \in \mathbb{K}$ of f is the coeffcient of $\textsc{lm}(f)$ in f; and the *leading term* of f is $\textsc{lt}(f) = \textsc{lc}(f) \cdot \textsc{lm}(f)$.

Lemma 19. *For every $f \in \mathbb{K}[\mathcal{A}]$ and $g \in \textsc{Mon}[\mathbb{K}]$ it holds $\textsc{lm}(g \cdot f) = g \cdot \textsc{lm}(f)$ and $\textsc{lt}(g \cdot f) = g \cdot \textsc{lt}(f)$.*

Proof. Due to Lemma 17, the $\preceq_{\text{lex}}$-largest monomial $\textsc{lm}(f)$ in f, after mutliplying f by a monomial g, is still the $\preceq_{\text{lex}}$-largest monomial in $g \cdot f$, namely $\textsc{lm}(g \cdot f) = g \cdot \textsc{lm}(f)$. As g is a monomial, we have $\textsc{lc}(g \cdot f) = \textsc{lc}(f)$, which implies $\textsc{lt}(g \cdot f) = g \cdot \textsc{lt}(f)$. ∎

Characteristic pairs. Instead of just monomials, we consider monomial-set pairs:

$$\mathrm{C}[\mathcal{A}] \ = \ \{(f, v) \in \textsc{Mon}[\mathcal{A}] \times \mathcal{P}_{\text{fin}}(A) \mid \textsc{vars}(f) \subseteq v\}.$$

Definition 20. The *characteristic pair* of f is defined as $\textsc{cp}(f) = (\textsc{lm}(f), \textsc{vars}(f)) \in \mathrm{C}[\mathcal{A}]$.

We extend division and lexicographic orders to $\mathrm{C}[\mathcal{A}]$ by combining them with inclusion (we overload symbols $\mid$ and $\preceq_{\text{lex}}$):

$$(f, v) \ \mid \ (f', v') \iff f \mid f' \ \wedge \ v \subseteq v'$$
$$(f, v) \prec_{\text{lex}} (f', v') \iff f \prec_{\text{lex}} f' \ \wedge \ v \subseteq v'.$$

While lexicographic order on monomials does not need to be well founded, its extension to $\mathrm{C}[\mathcal{A}]$ does. This is crucial in the sequel.

Lemma 21. *The extended $\preceq_{lex}$ is well founded.*

Proof. We need to show that every decreasing chain

$$(f_1, v_1) \succeq_{\text{lex}} (f_2, v_2) \succeq_{\text{lex}} \dots$$

eventually stabilises. The second component eventually stabilises, i.e., there is $n \in \mathbb{N}$ and $v \subseteq_{\text{fin}} A$ such that for all $i \geq n$ we have $v_i = v$. Since v is finite, lexicographic order is well founded on $\{f \in \textsc{Mon}[\mathcal{A}] \mid \textsc{vars}(f) \subseteq v\}$, and therefore $f_n \succeq_{\text{lex}} f_{n+1} \succeq_{\text{lex}} \dots$ eventually stabilises too. ∎

Similarly we extend $\sqsubseteq$ to $\mathrm{C}[\mathcal{A}]$, by combining relaxation of division order (3.5) with inclusion:

$$(m, v) \ \sqsubseteq \ (m', v') \iff \exists \iota \in \textsc{Emb}_{\mathcal{A}} : \iota(m) \mid m' \wedge \iota(v) \subseteq v'. \tag{3.6}$$

Lemma 22. *The extended $\sqsubseteq$ is a WQO, if $\mathcal{A}$ is ω-well structured.*

Proof. Given a pair $(f, v) \in \mathrm{C}[\mathcal{A}]$, define $(\ell_{(f,v)} : v \to \mathbb{N}) \in \textsc{Age}(\mathcal{A}, \mathbb{N})$ as follows:

$$\ell_{(f,v)}(a) \ = \ \begin{cases} \textsc{deg}_f\, a & \text{if } a \in \textsc{vars}(f) \\ 0 & \text{if } a \in v \setminus \textsc{vars}(f). \end{cases}$$

We observe that $(f, v) \sqsubseteq (f', v')$ exactly when $\ell_{(f,v)} \rightsquigarrow \ell_{(f',v')}$ are related by the quasi order determined by $\omega = (\mathbb{N}, \leq)$. Therefore $\sqsubseteq$ is a WQO, since $\mathcal{A}$ is ω-well structured. ∎

Proof of Theorem 13. Fix a totally ordered and ω-well structured domain $\mathcal{A} = (A, \preceq, \ldots)$. For simplicity, we restrict to the case when $\mathbb{K}$ is a *field* – the general proof for an arbitrary Noetherian commutative ring can be found in the full version [16]. We proceed by a careful adaptation of the classical proof, see e.g. [12, Section 2.5] using $\sqsubseteq$ in place of division order, and characteristic pairs in place of just leading monomials.

Consider an arbitrary ideal $\mathcal{I} \subseteq \mathbb{K}[\mathcal{A}]$, and the set of characteristic pairs of all polynomials in $\mathcal{I}$:

$$\textsc{cp}(\mathcal{I}) = \{\textsc{cp}(f) \,|\, f \in \mathcal{I}\}\,.$$

Let $J \subseteq \mathrm{C}[\mathcal{A}]$ be the upward closure of $\textsc{cp}(\mathcal{I})$ with respect to $\sqsubseteq$:

$$J = \{(f, v) \in \mathrm{C}[\mathcal{A}] \,|\, (f', v') \sqsubseteq (f, v) \text{ for some } (f', v') \in \textsc{cp}(\mathcal{I})\}\,.$$

The set J is thus upward-closed and therefore has a finite basis $B \subseteq_{\text{fin}} J$ (relying on Lemmas 5 and 22). W.l.o.g. $B \subseteq \textsc{cp}(\mathcal{I})$, as J is the upward closure of $\textsc{cp}(\mathcal{I})$. Therefore:

Claim 23. *For every $(f, v) \in B$, there is a polynomial $g_{f,v} \in \mathcal{I}$ such that $\textsc{cp}(g_{f,v}) = (f, v)$.*

Let $\mathcal{B} = \{g_{f,v} \,|\, (f, v) \in B\}$. We prove that $\mathcal{B}$ is a basis of $\mathcal{I}$, i.e., $\mathcal{I} = \langle\mathcal{B}\rangle$.

One inclusion is obvious, $\langle\mathcal{B}\rangle \subseteq \mathcal{I}$. In order to prove the opposite inclusion suppose, towards contradiction, that $\mathcal{I} \setminus \langle\mathcal{B}\rangle$ is nonempty. Take any polynomial $g \in \mathcal{I} \setminus \langle\mathcal{B}\rangle$ whose characteristic pair $(f, v) = \textsc{cp}(g)$ is $\preceq_{\text{lex}}$-minimal, i.e., there is no $g' \in \mathcal{I} \setminus \langle\mathcal{B}\rangle$ with $\textsc{cp}(g') \prec_{\text{lex}} \textsc{cp}(g)$ (cf. Lemma21). Choose any $(f', v') \in B$ such that $(f', v') \sqsubseteq (f, v)$, i.e., there is some $h \in \textsc{Mon}[\mathcal{A}]$ and $\iota \in \textsc{Emb}_{\mathcal{A}}$ such that

$$h \cdot \iota(f') = f \qquad \iota(v') \subseteq v.$$

Equivalently, using Claim 23 and taking $g' = g_{f',v'} \in \mathcal{B}$, we have:

$$h \cdot \iota(\textsc{lm}(g')) = \textsc{lm}(g) \qquad \iota(\textsc{vars}(g')) \subseteq \textsc{vars}(g).$$

As $\textsc{lm}(_)$ and $\textsc{vars}(_)$ both commute with the action of embeddings, the two equalities rewrite first to

$$h \cdot \textsc{lm}(\iota(g')) \;=\; \textsc{lm}(g) \qquad \textsc{vars}(\iota(g')) \;\subseteq\; \textsc{vars}(g), \tag{3.7}$$

and then, by Lemma 19, the first one rewrites further to

$$\textsc{lm}(h \cdot \iota(g')) \;=\; \textsc{lm}(g). \tag{3.8}$$

Let g'' be the polynomial appearing on the left-hand side: $g'' := h \cdot \iota(g')$. By (3.8), the leading terms appearing on both sides of (3.8) cancel out in $g - g''$. We thus have

$$\textsc{lm}(g - g'') \prec_{\text{lex}} \textsc{lm}(g). \tag{3.9}$$

Furthermore, by the first equality in (3.7) we get $\textsc{vars}(h) \subseteq \textsc{vars}(g)$, and then by the second one we get $\textsc{vars}(g'') \subseteq \textsc{vars}(g)$, and hence

$$\textsc{vars}(g - g'') \subseteq \textsc{vars}(g). \tag{3.10}$$

By (3.9) and (3.10) we thus derive:

$$\textsc{cp}(g - g'') \prec_{\text{lex}} \textsc{cp}(g). \tag{3.11}$$

Finally, as both $\langle\mathcal{B}\rangle$ and $\mathcal{I}$ are ideals, knowing that $g'' \in \langle\mathcal{B}\rangle \subseteq \mathcal{I}$ and $g \in \mathcal{I} \setminus \langle\mathcal{B}\rangle$, we deduce:

$$(g - g'') \in \mathcal{I} \setminus \langle\mathcal{B}\rangle. \tag{3.12}$$

Conditions (3.11) and (3.12) are in contradiction with $\preceq_{\text{lex}}$-minimality of $\textsc{cp}(g)$, which completes the proof of Theorem 13. ∎

Example 24. The base constructed in the proof of Theorem 13 consists of (representatives of) polynomials with $\sqsubseteq$-minimal characteristic pairs. For illustration, take ordered nonnegative integers as the domain of variables, $\mathcal{A} = \{a_0 < a_1 < ...\}$, and the rational field $\mathbb{Q}$. Consider an ideal $\mathcal{I} \subseteq \mathbb{Q}[\mathcal{A}]$ containing those polynomials without constant term, where the sum of coefficients of monomials of even degree is equal to the sum of coefficients of monomials of odd degree. An example of a base given by the proof of Theorem 13 is

$$\mathcal{B} = \left\{a_0^2 + a_0, a_0 - a_1\right\},$$

as the following two characteristic pairs are the $\sqsubseteq$-minimal elements in $\textsc{cp}(\mathcal{I})$:

$$c_1 = \textsc{cp}(a_0^2 + a_0) = (a_0^2, \{a_0\}) \qquad c_2 = \textsc{cp}(a_0 - a_1) = (a_1, \{a_0, a_1\}).$$

Indeed, consider the leading term of a non-zero polynomial $f \in \mathcal{I}$. If the degree of some variable in $\textsc{lm}(f)$ is at least 2, we have $c_1 \sqsubseteq \textsc{cp}(f)$. Otherwise all variables appearing in $\textsc{lm}(f)$ are of degree 1. As $f \in \mathcal{I}$, it uses more that one variable, and the largest of them must necessarily appear in $\textsc{lm}(f)$. Therefore $c_2 \sqsubseteq \textsc{cp}(f)$.

There are several other possible bases that could be chosen in the proof of Theorem 13, for instance $\mathcal{B}_n = \left\{a_0^2 + a_0, a_0^n + (-1)^n a_1\right\}$ for any $n \geq 1$. ◁

Proof of Theorem 12. Let $\mathcal{A} = (A, \ldots)$ be a relational structure and $\mathbb{K}$ a commutative ring such that $\mathbb{K}[\mathcal{A}]$ is equivariantly Noetherian. It is sufficient to prove that $(\textsc{Mon}[\mathcal{A}], \sqsubseteq)$ is a WQO, as it is isomorphic to the quasi order determined by $\omega = (\mathbb{N}, \leq)$ on $\textsc{Age}(\mathcal{A}, \mathbb{N} \setminus 0)$, and hence also on $\textsc{Age}(\mathcal{A}, \mathbb{N})$.

Given any infinite sequence of monomials $f_1, f_2, \ldots$, we prove that it contains a dominating pair. Let $\mathcal{I}_i \subseteq \mathbb{K}[\mathcal{A}]$ be the equivariant ideal generated by the set $\{f_1, \ldots, f_{i-1}\}$. Clearly, $\mathcal{I}_1 \subseteq \mathcal{I}_2 \subseteq \ldots$.

Claim 25. *$\mathcal{I}_i = \mathcal{I}_{i+1}$ for some $i \in \mathbb{N}$.*

To prove the claim, let $\mathcal{I} \subseteq \mathbb{K}[\mathcal{A}]$ be the equivariant ideal of polynomials generated by the infinite set $\{f_1, f_2, \ldots\}$. By assumption, it has a finite basis $\mathcal{B}$, $\mathcal{I} = \langle\mathcal{B}\rangle$. Choose $i \in \mathbb{N}$ such that $\mathcal{B} \subseteq \mathcal{I}_i$. Then $\mathcal{I}_i = \langle\mathcal{B}\rangle = \mathcal{I}$ and hence $\mathcal{I}_i = \mathcal{I}_{i+1} = \ldots = \mathcal{I}$.

By the claim we have $f_{i+1} \in \mathcal{I}_i$, i.e., for some $r_1, \ldots, r_i \in \mathbb{K}$, embeddings $\iota_1, \ldots, \iota_i \in \textsc{Emb}_{\mathcal{A}}$, and monomials $h_1, \ldots, h_i \in \textsc{Mon}[\mathbb{K}]$,

$$f_{i+1} \;=\; r_1 \cdot h_1 \cdot \iota_1(f_1) + \ldots + r_i \cdot h_i \cdot \iota_i(f_i).$$

Since f_{i+1} is a monomial, all monomials $h_k \cdot \iota_k(f_k)$ different than f_{i+1} necessarily cancel out, and moreover f_{i+1} has to be equal to $h_k \cdot \iota_k(f_k)$ for some $k \leq i$. In other words, we get a dominating pair $f_k \sqsubseteq f_{i+1}$, as required. ■

4. Application: Orbit-finitely generated vector spaces

We use Theorem 13 to give a simple proof of a generalisation of the key result of [6]: orbit-finitely generated vector spaces have finite chains of equivariant subspaces. In the following, $\mathbb{K}$ is a field and $\mathcal{A} = (A, \ldots)$ an ω-well structured relational structure.

Orbit-finitely generated vector spaces. For $d \in \mathbb{N}$, consider the vector space $\textsc{Lin}(A^d)$ freely generated by the set A^d of d-tuples of elements of A. Thus $\textsc{Lin}(A^d)$ consists of finite formal sums

$$r_1 \cdot t_1 + \ldots + r_m \cdot t_m, \tag{4.1}$$

where $r_1, \ldots, r_m \in \mathbb{K}$ and $t_1, \ldots, t_m \in A^d$. Equivalently, $\textsc{Lin}(A^d)$ may be defined as the set of all functions $v : A^d \to \mathbb{K}$ such that $v(t) \neq 0$ for finitely many $t \in A^d$. Addition of vectors (4.1) and multiplication by a scalar $r \in \mathbb{K}$ are then

defined pointwise, as expected. Embeddings $\iota \in \text{EMB}_{\mathcal{A}}$ act pointwise on tuples $t = (a_1, \ldots, a_d)$, namely $\iota(t) = (\iota(a_1), \ldots, \iota(a_d))$, and also pointwise on vectors:

$$\iota(r_1 \cdot t_1 + \ldots + r_m \cdot t_m) \;=\; r_1 \cdot \iota(t_1) + \ldots + r_m \cdot \iota(t_m).$$

A subset of $\text{LIN}(A^d)$ is called *equivariant* if it is invariant under action of $\text{EMB}_{\mathcal{A}}$.

Example 26. When $d = 2$, generators are ordered pairs $(a, a') \in A^2$. A vector in $\text{LIN}(A^2)$ can be presented as a finite directed graph, possibly containing self-loops, whose nodes are elements $a \in A$ and edges are labelled by elements of the field $\mathbb{K}$. Addition of vectors amounts to merging two such graphs, and adding labels of common edges. Multiplication by a scalar $r \in \mathbb{K}$ amount to multiplying labels of all edges by r. The set of those graphs where the sum of labels is 0 is an equivariant subspace of $\text{LIN}(A^2)$. ◁

We encode vectors as polynomials as follows. A single generator $t = (a_1, a_2, \ldots, a_d) \in A^d$ is encoded as the monomial

$$\text{ENC}(t) \;=\; {a_1}^1 \cdot {a_2}^2 \cdot \ldots \cdot {a_d}^{2^d}.$$

As repetitions $a_i = a_j$ are allowed, the degree of a variable a is thus equal to the sum of powers 2^i, ranging over positions i such that $a_i = a$. This guarantees injectivity of $\text{ENC}(_)$, namely different tuples t have different encodings. Then a vector $v = r_1 \cdot t_1 + \ldots + r_m \cdot t_m$ is encoded as the polynomial

$$\text{ENC}(v) \;=\; r_1 \cdot \text{ENC}(t_1) + \ldots + r_m \cdot \text{ENC}(t_m),$$

which yields a linear map $\text{ENC}(_) : \text{LIN}(A^d) \to \mathbb{K}[\mathcal{A}]$.

Vector space $\text{LIN}(A^d)$ is indeed *orbit-finitely generated* once $\mathcal{A}$ is assumed to be ω-well structured, due to the following lemma:

Lemma 27 (oligomorphicity[4])**.** *A^d is orbit-finite, namely $A^d = \text{EMB}_{\mathcal{A}}(B) \stackrel{def}{=} \{\iota(t) \mid \iota \in \text{EMB}_{\mathcal{A}}, t \in B\}$ for some $B \subseteq_{fin} A^d$.*

Any such finite set B we call a *presentation* of A^d.

Proof. Suppose A^d is orbit-infinite. In such case there is an infinite sequence $t_1, t_2, \ldots \in A^d$ such that $A^d = \text{EMB}_{\mathcal{A}}(\{t_1, t_2, \ldots\})$ and $(*)$ there is no $i, j \in \mathbb{N}$ and embedding $\iota \in \text{EMB}_{\mathcal{A}}$ such that $\iota(t_i) = t_j$. We know that $(\text{MON}[\mathbb{K}], \sqsubseteq)$ is a WQO and hence the infinite sequence of monomials

$$\text{ENC}(t_1),\ \text{ENC}(t_2),\ \ldots$$

in $\text{MON}[\mathbb{K}]$ admits a domination, namely $\iota(\text{ENC}(t_i)) \mid \text{ENC}(t_j)$ for some $i < j$ and $\iota \in \text{EMB}_{\mathcal{A}}$. A crucial fact is that all monomials used in encodings have the same degree $D = 1 + 2 + \ldots + 2^d = 2^{d+1} - 1$, understood as the sum of degrees of all variables. Therefore $\iota(\text{ENC}(t_i)) \mid \text{ENC}(t_j)$ implies $\iota(\text{ENC}(t_i)) = \text{ENC}(t_j)$. The encoding commutes with action of embeddings: $\iota(\text{ENC}(t)) = \text{ENC}(\iota(t))$. This implies $\text{ENC}(\iota(t_i)) = \text{ENC}(t_j)$, and by injectivity of $\text{ENC}(_)$ also $\iota(t_i) = t_j$, thus contradicting $(*)$. ■

[4]If we restrict to automorphisms, instead of all embeddings, this is a characterisation of oligomorphic automophism groups [9], and ω-categorical structures [20]. Furthermore, as mentioned in Section 2, when $\mathcal{A}$ is homogeneous, action of embeddings is essentially the same as action of automorphisms.

Finite length. Theorem 28 generalises [6, Lemma IV.8] which only considers $\mathcal{A} = \mathcal{Q} = (Q, \leq_{\mathcal{Q}})$. On the other hand, [6] provides explicite upper bound on the length.

Theorem 28 (Finite length)**.** *Let $\mathbb{K}$ be any field. If $\mathcal{A}$ is a reduct of an ω-well structured and totally ordered structure then strictly increasing chains of equivariant subspaces of $\textsc{Lin}(\mathcal{A}^d)$ are finite.*

Proof. Consider any increasing chain of equivariant subspaces

$$\mathcal{V}_1 \subseteq \mathcal{V}_2 \subseteq \dots$$

and let $\mathcal{I}_i$ be the ideal in $\mathbb{K}[\mathcal{A}]$ generated (in the classical sense) by $\textsc{enc}(\mathcal{V}_i)$, the encodings of vectors from $\mathcal{V}_i$. As action of embeddings commutes with encoding, equivariance of subspaces $\mathcal{V}_i$ implies equivariance of the sets $\textsc{enc}(\mathcal{V}_i)$. As action of embeddings commutes with addition and multiplication of polynomials, ideals $\mathcal{I}_i$ are equivariant. By Theorem 13 and Lemma 14, the chain $\mathcal{I}_1 \subseteq \mathcal{I}_2 \subseteq \dots$ stabilises, namely, $\mathcal{I}_i = \mathcal{I}_j$ for all sufficiently large i, j. We claim:

Claim 29. *$\mathcal{I}_i = \mathcal{I}_j$ implies $\textsc{enc}(\mathcal{V}_i) = \textsc{enc}(\mathcal{V}_j)$.*

Having the claim we immediately complete the proof, as $\textsc{enc}(\mathcal{V}_i) = \textsc{enc}(\mathcal{V}_j)$ implies $\mathcal{V}_i = \mathcal{V}_j$, by injectivity of $\textsc{enc}(_)$.

For demonstrating the claim it is sufficient to prove the inclusion $\textsc{enc}(\mathcal{V}_i) \supseteq \textsc{enc}(\mathcal{V}_j)$. Given an arbitrary polynomial $f \in \textsc{enc}(\mathcal{V}_j)$, we thus prove $f \in \textsc{enc}(\mathcal{V}_i)$. As $\mathcal{I}_i = \mathcal{I}_j$, we have $f \in \mathcal{I}_i$, i.e.,

$$f \;=\; r_1 \cdot h_1 \cdot \textsc{enc}(v_1) + \dots + r_n \cdot h_n \cdot \textsc{enc}(v_n) \tag{4.2}$$

for some vectors $v_1, \dots, v_n \in \mathcal{V}_i$, monomials $h_1, \dots, h_n \in \textsc{Mon}[\mathbb{K}]$ and ring elements $r_1, \dots, r_n \in \mathbb{K}$. As all monomials used in encodings have the same degree D, all monomials g appearing in $\textsc{enc}(v_1), \dots, \textsc{enc}(v_n)$ have degree $\textsc{deg}\, g = D$, as well as all monomials appearing in f. This means that all the monomials of degree larger than D, obtained as a multiplication $h_i \cdot g$, forcedly cancel out in the final sum (4.2). Formally ($\mathbf{0}$ is the zero polynomial):

$$\sum_{i \in I} r_i \cdot h_i \cdot \textsc{enc}(v_i) \;=\; \mathbf{0},$$

where the sum ranges over those i where h_i has positive degree: $I = \{i \in \{1 \dots n\} \mid \textsc{deg}\, h_i > 0\}$. All polynomials $r_i \cdot h_i \cdot \textsc{enc}(v_i)$, for $i \in I$, can be thus safely removed. As $\textsc{deg}\, h_i = 0$ exactly when h_i is a unit monomial, we end up with

$$f \;=\; \sum_{i \notin I} r_i \cdot \textsc{enc}(v_i) \;=\; \textsc{enc}(\sum_{i \notin I} r_i \cdot v_i),$$

the latter equality following from linearity of $\textsc{enc}(_)$. Therefore $f \in \textsc{enc}(\mathcal{V}_i)$, as required. ∎

Weighted Register Automata. Using Theorem 28 we obtain a generalisation of another result of [6]: decidability of zeroness of weighted register automata.

For $d \in \mathbb{N}$, let $Q_d = A^d \cup \{\bot\}$. The auxiliary element $\bot$ is assumed to be equivariant, namely $\iota(\bot) = \bot$ for all $\iota \in \textsc{Emb}_{\mathcal{A}}$. The vector space $\textsc{Lin}(Q_d)$ is defined like $\textsc{Lin}(A^d)$, as the set of finite formal sums of elements of Q_d.

Classically, transition function of a weighted automaton with finite state-space Q maps every input symbol and state to a vector over states: $\Sigma \times Q \to \textsc{Lin}(Q)$. Our definition of weighted d-register automaton (equivalent to the definition of [6]) is along the same lines, except that input alphabet is $\Sigma = A$, state-space is orbit-finite $Q = Q_d$, and transition function is equivariant.

A weighted d-register automaton consists of a *transition* function $\delta : A \times Q_d \to \textsc{Lin}(Q_d)$ and a *final* function $F : Q_d \to \mathbb{K}$, both equivariant, namely for every $a \in A$, $q \in Q_d$, and $\iota \in \textsc{Emb}_{\mathcal{A}}$:

$$\delta(\iota(a), \iota(q)) \;=\; \iota(\delta(a, q)) \qquad F(\iota(a)) \;=\; F(a).$$

F extends uniquely to a linear map $F : \textsc{Lin}(Q_d) \to \mathbb{K}$. Likewise, for every $a \in A$, the function $\delta(a, _) : Q_d \to \textsc{Lin}(Q_d)$ extends uniquely to a linear map $\widetilde{\delta}(a) : \textsc{Lin}(Q_d) \to \textsc{Lin}(Q_d)$. We overload $\perp$ and use it for vector defined by the singleton formal sum $\perp \in \textsc{Lin}(Q_d)$, the *initial* vector. The semantics of a weighted register automaton (d, δ, F) maps every word $w = a_1 \dots a_\ell \in A^*$ to the *final* vector

$$\widetilde{\delta}(w) \;=\; \widetilde{\delta}(a_\ell) \circ \dots \circ \widetilde{\delta}(a_1)(\perp) \;\in\; \textsc{Lin}(Q_d),$$

on which F is evaluated to get the *value* $F(\widetilde{\delta}(w)) \in \mathbb{K}$ assigned by the automaton to w. By equivariance of $\perp$, δ and F, the function $\widetilde{\delta}$ is equivariant and the value is stable under the pointwise action of $\textsc{Emb}_{\mathcal{A}}$ on words, $\iota(a_1 \dots a_\ell) = \iota(a_1) \dots \iota(a_\ell)$.

Lemma 30. *$F(\widetilde{\delta}(\iota(w))) = F(\widetilde{\delta}(w))$ for every $\iota \in \textsc{Emb}_{\mathcal{A}}$.*

The *zeroness* problem asks, given a weighted register automaton (d, δ, F), whether it is *zeroing*, i.e., whether it assigns value 0 to all input words. When an automaton is input to an algorithm, each of the functions δ, F is given by its values on a (naturally defined) presentation of its domain. We prove that zeroness is characterised by a finite testing set:

Theorem 31. *Let $\mathbb{K}$ be any field and $\mathcal{A}$ be a reduct of an ω-well structured and totally ordered structure. For every weighted register automaton there is a finite set $T \subseteq A^*$ such that the automaton is zeroing if and only if it assigs value 0 to all words in T.*

Proof. Consider a weighted register automaton (d, δ, F). For $i \in \mathbb{N}$, let $A^{\leq i} = A^0 \cup \dots \cup A^i$, and $\mathcal{V}_i \subseteq \textsc{Lin}(Q_d)$ be the vector space spanned by $V_i = \widetilde{\delta}(A^{\leq i}) \stackrel{\text{def}}{=} \left\{ \widetilde{\delta}(w) \,\middle|\, w \in A^{\leq i} \right\}$. Let B_i be a presentation of A^i, and let $B_{\leq i} = B_0 \cup \dots \cup B_i$. Therefore

$$A^{\leq i} = \textsc{Emb}_{\mathcal{A}}(B_{\leq i}), \tag{4.3}$$

and by equivariance of $\widetilde{\delta}$ we have:

$$V_i = \textsc{Emb}_{\mathcal{A}}(\widetilde{\delta}(B_{\leq i})). \tag{4.4}$$

Thus the sets V_i are equivariant, and so are the subspaces $\mathcal{V}_i$. As an immediate consequence of Theorem 28 we obtain:

Claim 32. *Strictly increasing chains of equivariant subspaces of $\textsc{Lin}(Q_d)$ are finite.*

Proof. We claim that subspaces of $\textsc{Lin}(Q_d)$ split into two types defined below. Consider an equivariant subspace $\mathcal{V} \subseteq \textsc{Lin}(Q_d)$ and its projection $\mathcal{V}' \subseteq \textsc{Lin}(A^d)$ to $\textsc{Lin}(A^d)$. If $\mathcal{V}$ contains a vector $r \cdot \perp$ for some $r \neq 0$ then $\mathcal{V}$ contains vectors $r \cdot \perp$ for all $r \in \mathbb{K}$ and therefore $\mathcal{V} = \mathcal{V}' \times \mathbb{K}$. Otherwise, every vector $v' \in \mathcal{V}'$ extends uniquely to a vector in $\mathcal{V}$. In both cases, whenever two subspaces $\mathcal{V}_1, \mathcal{V}_2 \subseteq \textsc{Lin}(Q_d)$ are related by strict inclusion, their projections $\mathcal{V}'_1, \mathcal{V}'_2$ to $\textsc{Lin}(A^d)$ also are:

$$\mathcal{V}_1 \subset \mathcal{V}_2 \implies \mathcal{V}'_1 \subset \mathcal{V}'_2. \tag{4.5}$$

Furthermore, a subspace of the first type is never included in a subspace of the second type. Therefore every strictly increasing chain $\mathcal{V}_1 \subset \mathcal{V}_2 \subset \dots$ of equivariant subspaces of $\textsc{Lin}(Q_d)$ splits into two subchains: a chain of subspaces of the second type, followed by chain of subspaces of the first type. By (4.5) each of the subchains

corresponds to a strictly increasing chain of subspaces of $\textsc{Lin}(A^d)$, and hence is finite by Theorem 28. In consequence, the whole chain is necessarily finite. ■

By the claim, the chain $\mathcal{V}_0 \subseteq \mathcal{V}_1 \subseteq \dots$, stabilises at some n:

$$\mathcal{V}_n = \mathcal{V}_{n+1} = \dots. \tag{4.6}$$

Since the transformation $\mathcal{V}_i \mapsto \mathcal{V}_{i+1}$ does not depend on i, the stabilisation (4.6) is guaranteed once $\mathcal{V}_n = \mathcal{V}_{n+1}$. Zeroness of (d, δ, F) is then equivalent to $F(\widetilde{\delta}(w)) = 0$ for all $w \in A^{\leq n}$. Using (4.3) and Lemma 30 we refine the characterisation:

Claim 33. *Zeroness of (d, δ, F) is equivalent to $F(\widetilde{\delta}(w)) = 0$ for all $w \in B_{\leq n}$.*

Hence $T = B_{\leq n}$ satisfies Theorem 31, which is thus proved. ■

Remark 34. Under mild computability assumptions on $\mathbb{K}$ and $\mathcal{A}$ (omitted here for simplicity, but satisfied by all ω-well structured domains mentioned so far), this characterisation yields decidability of the zeroness problem. This generalises [6, Lemma VIII.1], but without providing complexity upper bound. Indeed, a stabilisation point n (4.6) can be computed as follows: enumerate in parallel, for all $n \in \mathbb{N}$, linear combinations of vectors from V_n, and stop when

$$\widetilde{\delta}(B_{\leq n+1}) \;\subseteq\; V_n.$$

Indeed, by (4.4), this implies $V_{n+1} \subseteq V_n$, and hence $\mathcal{V}_{n+1} \subseteq \mathcal{V}_n$. Once a stabilisation point n is computed, zeroness of an automaton is decided by computing the value assigned to all words in $B_{\leq n}$.

5. Computability assumptions

We now formulate computability assumptions on a field $\mathbb{K}$ and structure $\mathcal{A}$ which are necessary in subsequent sections.

A field $\mathbb{K}$ is *computable*, i.e., its elements are finitely represented, and field operations can be computed using this representation. So is, for instance, the rational field $\mathbb{Q}$, as well as every finite field.

The rest of this section is devoted to spelling out computability assumptions on an ω-well structured structure $\mathcal{A} = (A, \dots)$, in Definition 38 below. Given $B \in \textsc{Age}(\mathcal{A})$, any restriction $\iota|B$ of an embedding to B we call *local embedding* of B. It is legitimate to use local embeddings instead of embeddings, as their action on a polynomial depends only on restriction to its variables:

Lemma 35. $\iota|_{\textsc{vars}(f)} = \iota'|_{\textsc{vars}(f)} \implies \iota(f) = \iota'(f)$.

By $\textsc{Emb}_{\mathcal{A}}|B$ we denote the set of all local embeddings of B:

$$\textsc{Emb}_{\mathcal{A}}|B \;=\; \{\iota|B \,|\, \iota \in \textsc{Emb}_{\mathcal{A}}\}.$$

For $B, B' \in \textsc{Age}(\mathcal{A})$, we consider the Cartesian product

$$B \boxtimes B' \;\stackrel{\text{def}}{=}\; (\textsc{Emb}_{\mathcal{A}}|B) \times (\textsc{Emb}_{\mathcal{A}}|B').$$

We extend to this set the action of embeddings: for $\kappa \in \textsc{Emb}_{\mathcal{A}}$ and $(\pi, \pi') \in B \boxtimes B'$, we put $\kappa(\pi, \pi') = (\kappa \circ \pi, \kappa \circ \pi')$. Clearly, $B \boxtimes B'$ is closed under this action, i.e., it is equivariant.

Lemma 36. *For $B, B' \in \textsc{Age}(\mathcal{A})$, $B \boxtimes B'$ is orbit-finite, namely $B \boxtimes B' = \textsc{Emb}_{\mathcal{A}}(F) \stackrel{def}{=} \{(\kappa \circ \pi, \kappa \circ \pi') \,|\, \kappa \in \textsc{Emb}_{\mathcal{A}}, (\pi, \pi') \in F\}$ for some finite $F \subseteq B \boxtimes B'$.*

Any such finite set F we call a *presentation* of $B \boxtimes B'$.

Proof. Let $n = |B|$ and $n' = |B'|$. The set $B \boxtimes B'$ is essentially a subset of $\mathcal{A}^{n+n'}$. Formally, having fixed enumerations $B = \{b_1, \dots, b_n\}$, $B' = \{b'_1, \dots, b'_{n'}\}$, we define an injective mapping $m : B \boxtimes B' \to \mathcal{A}^{n+n'}$ as expected:

$$(\pi, \pi') \;\mapsto\; (\pi(b_1), \dots, \pi(b_n), \pi'(b'_1), \dots, \pi'(b'_{n'})).$$

The mapping is equivariant, namely commutes with the action of embeddings: $m(\iota(\pi, \pi')) = \iota(m(\pi, \pi'))$ for every $\iota \in \mathrm{EMB}_{\mathcal{A}}$. Therefore, equivariance of $B \boxtimes B'$ implies equivariance of $m(B \boxtimes B')$. By Lemma 27, the set $\mathcal{A}^{n+n'}$ is orbit-finite. Importantly, the same proof shows that *every* equivariant subset of $\mathcal{A}^{n+n'}$ is orbit-finite. Therefore, $m(B \boxtimes B') = \mathrm{EMB}_{\mathcal{A}}(F)$ for some finite set F. We use injectivity and equivariance of m to deduce that $m^{-1}(F)$ is a finite presentation of $B \boxtimes B'$. ■

Example 37. For illustration, consider the structure $\omega = (\mathbb{N}, \leq)$, whose embeddings are strictly monotonic maps $\omega \to \omega$, and $B = \{2\}$ and $B' = \{1, 3\}$. A presentation of $B \boxtimes B'$ is given by F containing five pairs (π, π') of local embeddings, corresponding to five cases: $\pi(2)$ is smaller than $\pi'(1)$, or equal to $\pi'(1)$, or between $\pi'(1)$ and $\pi'(3)$, or equal to $\pi'(3)$, or larger than $\pi'(3)$:

$$\begin{array}{llll} \pi_1 : 2{\mapsto}2 & \pi'_1 : (1,3){\mapsto}(3,5) & \pi_4 : 2{\mapsto}3 & \pi'_4 : (1,3){\mapsto}(1,3) \\ \pi_2 : 2{\mapsto}2 & \pi'_2 : (1,3){\mapsto}(2,4) & \pi_5 : 2{\mapsto}4 & \pi'_5 : (1,3){\mapsto}(1,3) \\ \pi_3 : 2{\mapsto}2 & \pi'_3 : (1,3){\mapsto}(1,3) & & \end{array}$$

The following two pairs are not in F, for different reasons:

$$\sigma_1 : 2{\mapsto}4 \quad \sigma'_1 : (1,3){\mapsto}(1,4) \qquad \sigma_2 : 2{\mapsto}3 \quad \sigma'_2 : (1,3){\mapsto}(2,3).$$

The left pair (σ_1, σ'_1) belongs to $B \boxtimes B'$ but factorizes through (π_4, π'_4), namely $(\sigma_1, \sigma'_1) = \kappa(\pi_4, \pi'_4)$ where $\kappa(0) = 0, \kappa(1) = 1$, $\kappa(2) = 2$, and $\kappa(n) = n + 1$ for $n \geq 3$. The right pair does not even belong to $B \boxtimes B'$, as σ'_2 does not extend to an embedding. ◁

Definition 38. We say that an ω-well structured domain $\mathcal{A}$ is *computable* if its elements can be finitely represented, and the following tasks are computable using this representation:

- (C_1) Given $B \in \mathrm{AGE}(\mathcal{A})$ and $\pi : B \to A$, decide if $\pi \in \mathrm{EMB}_{\mathcal{A}}|B$.
- (C_2) Given $B, B' \in \mathrm{AGE}(\mathcal{A})$, compute a presentation of $B \boxtimes B'$.

In particular, every finite $\mathcal{A}$ is computable. ◁

(Our condition (C_2) is similar to EGB4 in [7].) By reusing the proof of [18, Proposition 3.4] we can show that the ordinal ω, seen as a relational structure, is computable. In fact the proof can be easily extended to ordinals α which are *effective*, by which we mean that elements of α (i.e., ordinals smaller than α) are finitely represented, and ordinal order $<$ and addition $+$ are computable using this representation. For example, the ordinals ω, ω^k, ω^ω and ε_0 are effective, using Cantor Normal Form.

Lemma 39. *Any effective ordinal α, seen as a relational structure, is computable.*

Proof. We start by considering ω; later we explain how to extend the proof to any effective ordinal. Let $B \subseteq_{\mathrm{fin}} \omega$ be a nonempty subset of ω. A function $\pi : B \to \omega$ extends to an embedding $\omega \to \omega$ if and only if the following conditions hold:

$$\begin{array}{ll} b \leq \pi(b) & \text{where } b = \min B \\ b_2 - b_1 \leq \pi(b_2) - \pi(b_1) & \text{for every } b_1 \leq b_2. \end{array} \tag{5.1}$$

This implies (C_1).

Concering (C_2), given $B, B' \in \mathrm{AGE}(\mathcal{A})$, we compute $B \boxtimes B'$ by a dynamic algorithm with respect to the orders on B and B': for every two initial segments

$C \subseteq B$ and $C' \subseteq B'$, we compute $C \boxtimes C'$, as follows (the algorithm is similar to [18, Proposition 3.4]).

The trivial set $\emptyset \boxtimes \emptyset = \{(\emptyset, \emptyset)\}$ contains just one pair of empty functions. Consider two initial segments $C \subseteq B$ and $C' \subseteq B'$. Assume $\overline{C} \boxtimes \overline{C}'$ is already computed for all pairs of initial segments smaller than C, C', i.e. for all pairs $(\overline{C}, \overline{C}') \neq (C, C')$ of initial segments $\overline{C} \subseteq C$ and $\overline{C}' \subseteq C'$. For computing $C \boxtimes C'$ we use $\overline{C} \subseteq C$ obtained from C by removing the largest element b (assuming C is nonempty), and $\overline{C}' \subseteq C'$ obtained from C' by removing the largest element b' (assuming C' is nonempty). We compute $C \boxtimes C'$ by applying the following three procedures (note that their pre-conditions are not disjoint):

- Suppose both C and C' are nonempty. For every $(\bar{\pi}, \bar{\pi}') \in \overline{C} \boxtimes \overline{C}'$ we extend $\bar{\pi}$ and $\bar{\pi}'$ to $\pi : C \to \omega$ and $\pi' : C' \to \omega$ by mapping both b and b' to the minimal value $\pi(b) = \pi'(b') \in \omega$ satisfying the condition (5.1) for both π and π'. We add all so obtained pairs (π, π') to $C \boxtimes C'$.
- Suppose C is nonempty (while C' is either empty or not). For every $(\bar{\pi}, \pi') \in \overline{C} \boxtimes C'$ we extend $\bar{\pi} : \overline{C} \to \omega$ to $\pi : C \to \omega$ by mapping b to the minimal value $\pi(b)$ outside of the range $\pi'(C')$ that satisfies the condition (5.1). We add all so obtained pairs (π, π') to $C \boxtimes C'$.
- Supposing C' is nonempty (while C is either empty or not), we proceed symmetrically to the previous case.

This completes the procedure to compute $B \boxtimes B'$. We have thus completed the proof for ω.

The proof extends to any ordinal α, as long as ordinal subtraction is computable in α, where subtraction of ordinals $\beta - \gamma$, for $\gamma < \beta < \alpha$, is defined as the unique ordinal η such that $\beta = \gamma + \eta$ (the order of addition is important since addition of ordinals is not commutative). For any effective ordinal α and $\gamma < \beta < \alpha$, subtraction $\beta - \gamma$ is computable by enumerating ordinals $\eta < \alpha$ and testing whether $\beta = \gamma + \eta$. ■

Lemma 40. *Lexicographic product of two computable domains is computable.*

Proof. Let $\mathcal{A} = (A, \ldots)$ and $\mathcal{B} = (B, \ldots)$ be two computable, ω-well structured domains. We will show that $\mathcal{A} \otimes \mathcal{B}$ is also computable.

First, assuming that $\mathcal{A}$ and $\mathcal{B}$ satisfy (C_1), we show that $\mathcal{A} \otimes \mathcal{B}$ satisfies (C_1) as well. For $B \in \textsc{Age}(\mathcal{A} \otimes \mathcal{B})$ let $\overline{B} \in \textsc{Age}(\mathcal{A})$ be its projection on the first coordinate. Moreover, for $a \in \overline{B}$, let $B_a = \{a' \in A' \mid (a, a') \in B\} \in \textsc{Age}(\mathcal{B})$. Given $B \in \textsc{Age}(\mathcal{A})$ and $\pi : B \to A \times A'$, we need to check if π extend to an embedding of $\mathcal{A} \otimes \mathcal{B}$. W.l.o.g. assume that π preserves equality in first coordinate since (this necessary condition can be easily checked). Then, π induces a map $\overline{\pi} : \overline{B} \to A$ and maps $\pi_a : B_a \to A'$ for $a \in \overline{B}$. By Lemma 8, π extends to an embedding of $\mathcal{A} \otimes \mathcal{B}$ only if $\overline{\pi}$ extends to an embedding $\bar{\iota}$ of $\mathcal{A}$ and for every $a \in \overline{B}$, the map π_a extend to an embedding ι_a of $\mathcal{B}$. Moreover if both these conditions are satisfied, then π does extend to the embedding of $\mathcal{A} \otimes \mathcal{B}$ determined by $\bar{\iota}$ and the family of embeddings $(\iota_a \in \textsc{Emb}_{\mathcal{B}})_{a \in A}$, where ι_a is arbitrary for $a \notin \overline{B}$. Since $\mathcal{A}$ and $\mathcal{B}$ satisfy (C_1), the conditions can be checked and hence $\mathcal{A} \otimes \mathcal{B}$ satisfies (C_1).

Now, assuming further that $\mathcal{A}$ and $\mathcal{B}$ satisfy (C_2), we show that $\mathcal{A} \otimes \mathcal{B}$ satisfies (C_2) as well, i.e., given $V, W \in \textsc{Age}(\mathcal{A} \otimes \mathcal{B})$, we show how to compute a presentation E of $V \boxtimes W$. Let $X \in \textsc{Age}(\mathcal{A})$ be the union of projections of V and W to the first coordinate:

$$X = \overline{V} \cup \overline{W} = \{a \in A \mid (a, b) \in V \cup W \text{ for some } b \in B\}.$$

Likewise, let $Y \in \textsc{Age}(\mathcal{B})$ be the union of projections of V and W to the second coordinate:

$$Y = \bigcup_{a \in X} (V_a \cup W_a) = \{b \in B \mid (a,b) \in V \cup W \text{ for some } a \in A\}.$$

Above we assume $V_a = \emptyset$ when $a \notin \overline{V}$, and likewise for W_a. Let $Z = X \times Y \in \textsc{Age}(\mathcal{A} \otimes \mathcal{B})$.

Claim 41. *If F is a presentation of $Z \boxtimes Z$, then*

$$E := \{(f|V, g|W) \mid (f,g) \in F\}$$

is a presentation of $V \boxtimes W$.

Proof: Consider $(p,q) \in V \boxtimes W$. There exists $(p',q') \in Z \boxtimes Z$ such that $p'|V = p$ and $q'|W = q$. There exists $(f,g) \in F$ and $\iota \in \textsc{Emb}_{\mathcal{A}\otimes\mathcal{B}}$ such that

$$\iota \circ f = p' \qquad \text{and} \qquad \iota \circ g = q' \ .$$

Which implies

$$\iota \circ f|V = p \qquad \text{and} \qquad \iota \circ g|W = q \ .$$

As $(p,q) \in V \boxtimes W$ is arbitrary, the claim is proved. ◁

It remains to show how to compute a presentation F of $Z \boxtimes Z$. As $\mathcal{A}$ and $\mathcal{B}$ satisfy ($\textsc{c}_2$), we can compute presentations S and T of $X \boxtimes X$ and $Y \boxtimes Y$ respectively. Before progressing further we notice that for any set U we have $U \boxtimes \emptyset = \{(\mathrm{Id}_U, \mathrm{Id}_\emptyset)\}$ and $\emptyset \boxtimes U = \{(\mathrm{Id}_\emptyset, \mathrm{Id}_U)\}$. Let $s = (s_1, s_2)$ be an arbitrary element in S. Let $s(X) := s_1(X) \cup s_2(X)$. We define a family of finite subsets I_s indexed by $s(X)$ as follows:

$$I_s(a) = \begin{cases} T & \text{when } a \in s_1(X) \cap s_2(X) \\ \{(\mathrm{Id}_Y, \mathrm{Id}_\emptyset)\} & \text{when } a \in s_1(X) \setminus s_2(X) \\ \{(\mathrm{Id}_\emptyset, \mathrm{Id}_Y)\} & \text{when } a \in s_2(X) \setminus s_1(X). \end{cases}$$

We put:

$$I_s := \prod_{a \in s(X)} I_s(a).$$

For every $t = (t_1^a, t_2^a)_{a \in s(X)} \in I_s$ we define $f^{s,t} = (f_1^{s,t}, f_2^{s,t}) \in Z \boxtimes Z$ as:

$$f_i^{s,t}(a,b) = (s_i(a), t_i^{s_i(a)}(b)).$$

We now prove that

$$F := \left\{ f^{s,t} \,\middle|\, s \in S,\ t \in I_s \right\}$$

is a presentation of $Z \boxtimes Z$. Towards this aim, consider an arbitrary $u = (u_1, u_2) \in Z \boxtimes Z$. The two local embeddings u_1, u_2 induce maps $\overline{u}_1, \overline{u}_2 \in \textsc{Emb}_{\mathcal{A}}|X$ and hence $(\overline{u}_1, \overline{u}_2) \in X \boxtimes X$. Since S is a presentation of $X \boxtimes X$, there exists $\overline{j} \in \textsc{Emb}_{\mathcal{A}}$ and $s = (s_1, s_2) \in S$ such that

$$\overline{j} \circ s_1 = \overline{u}_1 \qquad \text{and} \qquad \overline{j} \circ s_2 = \overline{u}_2 \ .$$

This implies the following equalities:

$$\overline{j}(s_1(X) \cap s_2(X)) = \overline{u}_1(X) \cap \overline{u}_2(X)$$
$$\overline{j}(s_1(X) \setminus s_2(X)) = \overline{u}_1(X) \setminus \overline{u}_2(X)$$
$$\overline{j}(s_2(X) \setminus s_1(X)) = \overline{u}_2(X) \setminus \overline{u}_1(X).$$

For every $a \in s(X)$, the local embeddings u_1 and u_2 induce local embeddings u_1^a and u_2^a in $\textsc{Emb}_{\mathcal{B}}|Y$ and $\textsc{Emb}_{\mathcal{B}}|Y$, respectively. Since T is a presentation of $Y \boxtimes Y$, there exists $(p_1^a, p_2^a) \in T$ and $j_a \in \textsc{Emb}_{\mathcal{B}}$ such that

$$j_a \circ p_1^a = u_1^a \qquad \text{and} \qquad j_a \circ p_2^a = u_2^a \ .$$

Define $t \in I_s$ as

$$(t_1^a, t_2^a) = \begin{cases} (p_1^a, p_2^a) & \text{when } a \in s_1(X) \cap s_2(X) \\ (\mathrm{Id}_Y, \mathrm{Id}_\emptyset) & \text{when } a \in s_1(X) \setminus s_2(X) \\ (\mathrm{Id}_\emptyset, \mathrm{Id}_Y) & \text{when } a \in s_2(X) \setminus s_1(X) \ . \end{cases}$$

We finish the proof by showing that there exists $\iota \in \textsc{Emb}_{\mathcal{A}\otimes\mathcal{B}}$ such that

$$\iota \circ f_1^{s,t} = u_1 \qquad \text{and} \qquad \iota \circ f_2^{s,t} = u_2 \ . \tag{5.2}$$

For $a \in s(X)$ there exists $v_1^a, v_2^a \in \textsc{Emb}_{\mathcal{B}}$ such that

$$v_1^a | X = u_1^a \qquad \text{and} \qquad v_2^a | X = u_2^a \ .$$

Define a family of embeddings ι_a indexed by $a \in \mathcal{A}$ as

$$\iota_a = \begin{cases} j_a & \text{when } a \in s_1(X) \cap s_2(X) \\ v_1^a & \text{when } a \in s_1(X) \setminus s_2(X) \\ v_2^a & \text{when } a \in s_2(X) \setminus s_1(X) \\ \mathrm{Id}_Y & \text{otherwise.} \end{cases}$$

Using Lemma 8 we can determine $\iota \in \textsc{Emb}_{\mathcal{A}\otimes\mathcal{B}}$ by $\overline{j}$ and the family of embeddings ι_a. It is routine to verify that ι satisfies (5.2). ■

6. Gröbner bases

In this section we set up the background for deciding the ideal membership problem in Section 7. Let $\mathbb{K}$ be a computable field. Let $\mathcal{A} = (A, \preceq, \ldots)$ be a computable, ω-well structured and *well ordered* domain (we thus strengthen the total order assumption). Therefore, the lexicographic order on monomials is well founded.

Division step. Let $\mathcal{G} \subseteq \mathbb{K}[\mathcal{A}] \setminus \{\mathbf{0}\}$ be a set of nonzero polynomials. Given polynomials $f, f' \in \mathbb{K}[\mathcal{A}]$, we write

$$f \longrightarrow_{\mathcal{G}} f' \tag{6.1}$$

if f' is the remainder of *division* of f by some polynomial $\iota(g)$ from $\textsc{Emb}_{\mathcal{A}}(\mathcal{G})$. Formally, we define (6.1) to mean that

$$f = r \cdot h \cdot \iota(g) + f' \tag{6.2}$$

for some $r \in \mathbb{K}, h \in \textsc{Mon}[\mathcal{A}]$, $\iota \in \textsc{Emb}_{\mathcal{A}}$ and $g \in \mathcal{G}$ such that the leading monomial $\textsc{lm}(r \cdot h \cdot \iota(g))$ (equal to $h \cdot \iota(\textsc{lm}(g))$) appears in f with the same coefficient as in $r \cdot h \cdot \iota(g)$ (namely $r \cdot \textsc{lc}(g)$). The monomial $\textsc{lm}(r \cdot h \cdot \iota(g))$ is called the *head* of the division step (6.1). The head is the largest monomial that disappears from f, i.e., does not appear any more in $f' = f - r \cdot h \cdot \iota(g)$. All monomials in f lexicographically larger than the head appear also in f' with the same coefficients, while some monomials lexicographically smaller than the head may be added or removed by a division step.

Recall that $\langle \mathcal{G} \rangle$ denotes the equivariant ideal generated by $\mathcal{G}$. Using equality (6.2) we deduce:

Lemma 42. *Division step preserves membeship in $\langle \mathcal{G} \rangle$: if $f \longrightarrow_{\mathcal{G}} f'$ then $f \in \langle \mathcal{G} \rangle \iff f' \in \langle \mathcal{G} \rangle$.*

A polynomial f is said to be *reduced* with respect to $\mathcal{G}$ if there is no $f' \in \mathbb{K}[\mathcal{A}]$ such that $f \longrightarrow_{\mathcal{G}} f'$.

Lemma 43. *Let $\mathcal{G} \subseteq \mathbb{K}[\mathcal{A}]$ be a finite set of nonzero polynomials. Every sequence of division steps*

$$f_1 \longrightarrow_{\mathcal{G}} f_2 \longrightarrow_{\mathcal{G}} f_3 \longrightarrow_{\mathcal{G}} \dots \tag{6.3}$$

terminates, i.e., ends in a reduced polynomial.

Proof. By *reduction ab absurdum*: assuming an infinite sequence of division steps (6.3), we deduce that the lexicographic order $\preceq_{\text{lex}}$ on monomials $\textsc{Mon}[\mathcal{A}]$ is not well founded.

Given an infinite sequence of division steps (6.3), we construct an infinite forest $\mathcal{F}$, whose nodes are labelled by monomials, as the limit of an increasing sequence of finite forests $\mathcal{F}_1, \mathcal{F}_2, \dots$. The construction will maintain the invariant that all monomials appearing in f_i are among labels of leaves in $\mathcal{F}_i$ (we *mark* these leaves, for distinguishing them from other leaves). The forest $\mathcal{F}_1$ consists just of (finitely many) roots, labeled by monomials appearing in f_1, all of them marked. Given $\mathcal{F}_i$, we define $\mathcal{F}_{i+1}$ by adding new children to the marked leaf v labelled by the head monomial m of the division step $f_1 \longrightarrow_{\mathcal{G}} f_{i+1}$. The leaves are labelled by monomials that are added by the division step, and all are marked. The node v itself becomes unmarked (even if it remains a leaf), and likewise do all other leaves labelled by monomials of f_i that do not appear in f_{i+1}.

By the construction of $\mathcal{F}$, every two monomials m, m' related by an edge are in strict lexicographic order: $m \succ_{\text{lex}} m'$. As the sequence (6.3) is infinite, the forest is also infinite and, by König's lemma, forcedly contains an infinite branch. This implies existence of an infinite strictly decreasing chain of monomials with respect to lexicographic order, which is a contradiction. ■

We write $f \longrightarrow^*_{\mathcal{G}} f'$ if there is some finite sequence of division steps from f to f' and f' is reduced.

Gröbner basis. A set $\mathcal{G} \subseteq \mathbb{K}[\mathcal{A}]$ is said to be a *Gröbner basis* if for every $f \in \langle \mathcal{G} \rangle$ there exists $g \in \mathcal{G}$ such that $\textsc{lm}(g) \sqsubseteq \textsc{lm}(f)$.

Lemma 44. *If $\mathcal{G} \subseteq \mathbb{K}[\mathcal{A}]$ is a Gröbner basis then for every polynomial $f \in \mathbb{K}[\mathcal{A}]$, the following conditions are equivalent: (1) $f \in \langle \mathcal{G} \rangle$; (2) $f \longrightarrow^*_{\mathcal{G}} \mathbf{0}$; (3) every sequence $f \longrightarrow^*_{\mathcal{G}} f'$ ends in $f' = \mathbf{0}$.*

Proof. The implication (2) $\Longrightarrow$ (1) holds for any $\mathcal{G}$. Indeed, a sequence $f \longrightarrow^*_{\mathcal{G}} \mathbf{0}$ of division steps yields a decomposition of f into a sum:

$$f \;=\; r_1 \cdot h_1 \cdot \iota_1(g_1) \;+\; \dots \;+\; r_n \cdot h_n \cdot \iota_n(g_n),$$

where $r_1, \dots, r_n \in \mathbb{K}$, $g_1, \dots, g_n \in \mathcal{G}$, $h_1, \dots, h_n \in \mathbb{K}[\mathcal{A}]$ and $\iota_1, \dots, \iota_n \in \textsc{Emb}_{\mathcal{A}}$, which proves membership of f in $\langle \mathcal{G} \rangle$.

For (1) $\Longrightarrow$ (3), suppose $f \in \langle \mathcal{G} \rangle$, and consider any sequence of division steps starting in f. By Lemma 43 the sequence is necessarily finite, and hence ends in some reduced polynomial f'. By Lemma 42, $f' \in \langle \mathcal{G} \rangle$. However, as $\mathcal{G}$ is a Gröbner basis, no nonzero polynomial in $\langle \mathcal{G} \rangle$ is reduced, and hence $f' = \mathbf{0}$.

The last implication (3) $\Longrightarrow$ (2) holds vacuously, as every polynomial f admits some sequence of division steps $f \longrightarrow^*_{\mathcal{G}} f'$. ■

Remark 45. Note that the proof of Theorem 13 shows that every equivariant ideal has a finite Gröbner basis. In the sequel we provide a way to *compute* such a basis.

Buchberger's algorithm. By the least common multiple of two monomials $f, f' \in \textsc{Mon}[\mathcal{A}]$, denoted as $\textsc{lcm}(f, f')$, we mean the least (with respect to divisibility) monomial g divisible by both f and f', i.e. satisfying $f \mid g$ and $f' \mid g$.

Definition 46 (S-polynomials)**.** Given two distinct polynomials $f, g \in \mathbb{K}[\mathcal{A}]$, $f \neq g$, we define their *S-polynomial* $\mathcal{S}(f,g)$ as

$$\mathcal{S}(f,g) \;:=\; \left(\frac{h}{\mathrm{LT}(f)} \cdot f\right) - \left(\frac{h}{\mathrm{LT}(g)} \cdot g\right),$$

where $h = \mathrm{LCM}(\mathrm{LM}(f), \mathrm{LM}(g))$. We say that monomial h is *cancelled out* in $\mathcal{S}(f,g)$. For a subset $\mathcal{G} \subseteq \mathbb{K}[\mathcal{A}]$ of polynomials, we put:

$$\mathcal{S}_{\mathcal{G}} \;=\; \bigcup_{(f,g)\in \mathrm{EMB}_{\mathcal{A}}(\mathcal{G}), f\neq g} \mathcal{S}(f,g).$$

Expanding $\mathrm{EMB}_{\mathcal{A}}(\mathcal{G}) = \{\iota(g) \,|\, \iota \in \mathrm{EMB}_{\mathcal{A}},\ g \in \mathcal{G}\}$, we get:

$$\mathcal{S}_{\mathcal{G}} \;=\; \{\mathcal{S}(\iota(f), \kappa(g)) \,|\, \iota, \kappa \in \mathrm{EMB}_{\mathcal{A}},\ f, g \in \mathcal{G},\ \iota(f) \neq \kappa(g)\}\,.$$

As action of embeddings commutes with S-polynomials, namely $\mathcal{S}(\kappa(f), \kappa(g)) = \kappa(\mathcal{S}(f,g))$ for every $\kappa \in \mathrm{EMB}_{\mathcal{A}}$ and $f, g \in \mathbb{K}[\mathcal{A}]$, the set $\mathcal{S}_{\mathcal{G}}$ is closed under the action of $\mathrm{EMB}_{\mathcal{A}}$, i.e. is equivariant: $\mathcal{S}_{\mathcal{G}} = \mathrm{EMB}_{\mathcal{A}}(\mathcal{S}_{\mathcal{G}})$. Using assumption ($\mathrm{C}_2$) we derive the following key fact:

Lemma 47 (Computable presentation)**.** *For every finite set of polynomials $\mathcal{G} \subseteq \mathbb{K}[\mathcal{A}]$, the set $\mathcal{S}_{\mathcal{G}}$ is orbit-finite, namely*

$$\mathcal{S}_{\mathcal{G}} \;=\; \mathrm{EMB}_{\mathcal{A}}(\mathcal{P}_{\mathcal{G}}) \;\stackrel{def}{=}\; \{\iota(f) \,|\, \iota \in \mathrm{EMB}_{\mathcal{A}}, f \in \mathcal{P}_{\mathcal{G}}\}$$

for some presentation $\mathcal{P}_{\mathcal{G}} \subseteq_{fin} \mathcal{S}_{\mathcal{G}}$, which is moreover computable.

Proof. We start by showing orbit-finiteness of $\mathcal{S}_{\mathcal{G}}$. As $\mathcal{G}$ is finite and $\mathcal{S}_{\mathcal{G}} = \bigcup_{g,g'\in\mathcal{G}} \mathcal{S}_{g,g'}$, where

$$\mathcal{S}_{g,g'} \;=\; \{\mathcal{S}(\iota(g), \iota'(g')) \,|\, \iota, \iota' \in \mathrm{EMB}_{\mathcal{A}},\ \iota(g) \neq \iota'(g')\}\,,$$

it is enough to prove that $\mathcal{S}_{g,g'}$ is orbit-finite for every two fixed polynomials $g, g' \in \mathcal{G}$. Let $B = \mathrm{VARS}(g)$ and $B' = \mathrm{VARS}(g')$, and let $F_{g,g'}$ be a finite presentation of $B \boxtimes B'$ (recall Lemma 36). Relying on Lemma 35, for a local embedding $\pi \in \mathrm{EMB}_{\mathcal{A}}|B$ we write $\pi(g)$ to mean the renaming of g by *any* extension of π to an embedding of $\mathcal{A}$. Likewise we write $\pi'(g')$, for $\pi' \in \mathrm{EMB}_{\mathcal{A}}|B'$:

$$\mathcal{S}_{g,g'} \;=\; \{\mathcal{S}(\pi(g), \pi'(g')) \,|\, (\pi, \pi') \in B \boxtimes B',\ \pi(g) \neq \pi'(g')\}\,.$$

Finally, we define $\mathcal{P}_{g,g'} \subseteq \mathcal{S}_{g,g'}$ as the subset of $\mathcal{S}_{g,g'}$ obtained by restricting to local pairs of embeddings (π, π') belonging to $F_{g,g'}$:

$$\mathcal{P}_{g,g'} \;=\; \{\mathcal{S}(\pi(g), \pi'(g')) \,|\, (\pi, \pi') \in F_{g,g'},\ \pi(g) \neq \pi'(g')\}\,.$$

As F is finite, $\mathcal{P}_{g,g'}$ is finite as well, and orbit-finiteness of $\mathcal{S}_{g,g'}$ follows once we prove that $\mathcal{P}_{g,g'}$ is a presentation of $\mathcal{S}_{g,g'}$:

Claim 48. $\mathcal{S}_{g,g'} = \mathrm{EMB}_{\mathcal{A}}(\mathcal{P}_{g,g'})$.

Proof: For proving the inclusion $\mathrm{EMB}_{\mathcal{A}}(\mathcal{P}_{g,g'}) \subseteq \mathcal{S}_{g,g'}$, we take any polynomial $\kappa(f) \in \mathrm{EMB}_{\mathcal{A}}(\mathcal{P}_{g,g'})$, where $\kappa \in \mathrm{EMB}_{\mathcal{A}}$ and $f = \mathcal{S}(\pi(g), \pi'(g')) \in \mathcal{P}_{g,g'}$, and argue that $\kappa(f) \in \mathcal{S}_{g,g'}$. As action of embeddings commutes with S-polynomials, we have

$$\kappa(\mathcal{S}(\pi(g), \pi'(g'))) \;=\; \mathcal{S}(\kappa(\pi(g)), \kappa(\pi'(g'))).$$

As κ is injective, knowing $\pi(g) \neq \pi'(g')$, we also know $\kappa(\pi(g)) \neq \kappa(\pi'(g'))$, and therefore $\kappa(f) \in \mathcal{S}_{g,g'}$.

The opposite inclusion $\mathcal{S}_{g,g'} \subseteq \mathrm{EMB}_{\mathcal{A}}(\mathcal{P}_{g,g'})$ is shown similarly. We take any polynomial $\mathcal{S}(\pi(g), \pi'(g')) \in \mathcal{S}_{g,g'}$. As $F_{g,g'}$ is a presentation of $B \boxtimes B'$, there is some $(\overline{\pi}, \overline{\pi}') \in F_{g,g'}$ and $\kappa \in \mathrm{EMB}_{\mathcal{A}}$ such that $(\pi, \pi') = (\kappa \circ \overline{\pi}, \kappa \circ \overline{\pi}')$. Knowing $\kappa(\pi(g)) \neq \kappa(\pi'(g'))$, we also know $\pi(g) \neq \pi'(g')$, and therefore $\mathcal{S}(\overline{\pi}(g), \overline{\pi}'(g')) \in \mathcal{P}_{g,g'}$. As action of embeddings commutes with S-polynomials, we get

$$\mathcal{S}(\pi(g), \pi'(g')) \;=\; \kappa(\mathcal{S}(\overline{\pi}(g), \overline{\pi}'(g'))),$$

and in consequence $\mathcal{S}(\pi(g), \pi'(g')) \in \mathrm{EMB}_{\mathcal{A}}(\mathcal{P}_{g,g'})$. ◁

Union of all the sets $\mathcal{P}_{g,g'}$ yields therefore a presentation of $\mathcal{S}_{\mathcal{G}}$:

$$\mathcal{P}_{\mathcal{G}} \stackrel{\text{def}}{=} \bigcup_{g,g' \in \mathcal{G}} \mathcal{P}_{g,g'}.$$

It is moreover computable: for every $g, g' \in \mathcal{G}$ the finite set $F_{g,g'}$ is computable, by assumption (C_2), and therefore the set $\mathcal{P}_{g,g'}$ is computable as well. ■

Lemma 49 (Orbit-finite Buchberger criterion)**.** *A finite set $\mathcal{G} \subseteq \mathbb{K}[\mathcal{A}]$ is a Gröbner basis if and only if for all $h \in \mathcal{S}_{\mathcal{G}}$,*

$$h \longrightarrow_{\mathcal{G}}^{*} \mathbf{0}.$$

Proof. According to our definition, a finite set $\mathcal{G} \subseteq \mathbb{K}[\mathcal{A}]$ of polynomials is a Gröbner basis if and only if the *orbit-finite* set $\mathrm{EMB}_{\mathcal{A}}(\mathcal{G})$ is a Gröbner basis in the classical sense. As the set $\mathcal{S}_{\mathcal{G}}$ contains exactly S-polynomials of pairs of polynomials from $\mathrm{EMB}_{\mathcal{A}}(\mathcal{G})$, we reuse the proof of classical Buchberger criterion (which is still valid for infinite sets, see e.g. [12, Chapter 2]) to deduce the lemma. (This is actually the only step which requires the stengthening of assumptions, i.e., well ordered $\mathcal{A}$.) ■

Lemma 50 (Finite Buchberger criterion)**.** *A finite set $\mathcal{G} \subseteq \mathbb{K}[\mathcal{A}]$ is a Gröbner basis if and only if for all $h \in \mathcal{P}_{\mathcal{G}}$,*

$$h \longrightarrow_{\mathcal{G}}^{*} \mathbf{0}.$$

Proof. The 'only if' direction follows immediately by Lemma 49. For the 'if' direction, suppose that $h \longrightarrow_{\mathcal{G}} \mathbf{0}$ for all $h \in \mathcal{P}_{\mathcal{G}}$. Consider any $h' \in \mathcal{S}_{\mathcal{G}}$, i.e. $h' = \iota(h)$ for some $\iota \in \mathrm{EMB}_{\mathcal{A}}$ and $h \in \mathcal{P}_{\mathcal{G}}$. By assumption, $h \longrightarrow_{\mathcal{G}}^{*} \mathbf{0}$. We observe that the action of $\mathrm{EMB}_{\mathcal{A}}$ (i.e., renaming of variables) preserves division steps, namely

$$f \longrightarrow_{\mathcal{G}} f' \implies \iota(f) \longrightarrow_{\mathcal{G}} \iota(f'),$$

and therefore $\iota(h) \longrightarrow_{\mathcal{G}}^{*} \iota(\mathbf{0}) = \mathbf{0}$. As $h' \in \mathcal{S}_{\mathcal{G}}$ is arbitrary, using Lemma 49 we conclude that $\mathcal{G}$ is a Gröbner basis. ■

Algorithm 1 is an adaptation of the classical Buchberger algorithm to our setting. The adaptation amounts to using only polynomials from $\mathcal{P}_{\mathcal{G}}$ instead of all S-polynomials.

Algorithm 1 EQUIVARIANT BUCHBERGER ALGORITHM

Input: A finite set of polynomials $\mathcal{G} \subseteq_{\text{fin}} \mathbb{K}[\mathcal{A}]$.

repeat
  for $f \in \mathcal{P}_{\mathcal{G}}$ **do**
    compute some reduced $h \in \mathbb{K}[\mathcal{A}]$ such that $f \longrightarrow_{\mathcal{G}}^{*} h$
    if $h \neq \mathbf{0}$ **then** $\mathcal{G} \leftarrow \mathcal{G} \cup \{h\}$
until $\mathcal{G}$ stabilizes

Output: $\mathcal{G}$

The above description is indeed an algorithm as, relying on our computability assumption (C_1), we prove that one can compute some division step from a given polynomial f:

Lemma 51 (Computability)**.** *Given $f \in \mathbb{K}[\mathcal{A}]$ and $\mathcal{G} \subseteq_{fin} \mathbb{K}[\mathcal{A}]$, one can check if f is reduced wrt. $\mathcal{G}$ and, if f is not, compute some $f' \in \mathbb{K}[\mathcal{A}]$ such that $f \longrightarrow_{\mathcal{G}} f'$.*

Proof. For every polynomial $g \in \mathcal{G}$ and monomial m of f, we compute the set $D_{g,m}$ of those functions $\pi : \textsc{vars}(\textsc{lm}(g)) \to \textsc{vars}(m)$ that extend to an embedding in $\textsc{Emb}_{\mathcal{A}}$ (which is checked using assumption ($\textsc{c}_1$)) and satisfy $\pi(\textsc{lm}(g)) \mid m$. If $D_{g,m} = \emptyset$ for all g and m, the polynomial f is reduced wrt. $\mathcal{G}$. Otherwise we pick up arbitrary function $\pi \in D_{g,m}$, and compute some its extension $\pi' : \textsc{vars}(g) \to A$ to $\textsc{vars}(g)$ that still extends to an embedding in $\textsc{Emb}_{\mathcal{A}}$. This is doable by enumerating all (countably many) candidates $\pi' : \textsc{vars}(g) \to A$ and checking them (again relying on assumption ($\textsc{c}_1$)). Termination is guaranteed as π does extend to an embedding. Finally, we divide m by $\pi'(\textsc{lm}(g))$ which yields a monomial h satisfying $m = h{\cdot}\pi'(\textsc{lm}(g))$, and compute the remainder $f' = f - r{\cdot}\textsc{lc}(g)^{-1}{\cdot}h{\cdot}\pi'(g)$, where $r \in \mathbb{K}$ is the coefficient of m in f. ∎

The algorithm is nondeterministic, namely it may choose different sequences of division steps $f \longrightarrow^*_{\mathcal{G}} h$ and hence different reduced polynomials h, and therefore it may have many different runs for the same input $\mathcal{G}$. Nevertheless, we prove termination of its every run, and correctness of output.

Lemma 52 (Termination)**.** *Every run of Algorithm 1 terminates on every input* $\mathcal{G} \subseteq \mathbb{K}[\mathcal{A}]$.

Proof. Towards contradiction, suppose some run of the algorithm is infinite. Let $\mathcal{G}_n \subseteq \mathbb{K}[\mathcal{A}]$ be the value of $\mathcal{G}$ after n iterations of the while loop. Let $\mathcal{M}_n = \{\textsc{lm}(g) \,|\, g \in \mathcal{G}_n\} \subseteq \textsc{Mon}[\mathcal{A}]$. For every n we have $\mathcal{M}_n \subseteq \mathcal{M}_{n+1}$. As every polynomial h added in $(n+1)$th iteration is reduced with respect to $\mathcal{G}_n$, we have $\textsc{lm}(h) \notin \mathcal{M}_n$, and hence $\mathcal{M}_n \subsetneq \mathcal{M}_{n+1}$ for every n. Therefore we can form an infinite sequence by picking up, for every n, a monomial $m_n \in \mathcal{M}_{n+1} \setminus \mathcal{M}_n$. This sequence is *bad*, i.e. $m_i \not\sqsubseteq m_j$ for every $i < j$, which is in contradiction with $\sqsubseteq$ being a WQO. ∎

Lemma 53 (Correctness)**.** *Every run of Algorithm 1 computes, given an input* $\mathcal{G} \subseteq \mathbb{K}[\mathcal{A}]$*, a Gröbner basis of* $\langle \mathcal{G} \rangle$.

Proof. Each polynomial h added to $\mathcal{G}$ in the algorithm is obtained from some element of $\mathcal{G}$ by a sequence of division steps $\longrightarrow_{\mathcal{G}}$, and hence belongs to $\langle \mathcal{G} \rangle$. Therefore $\langle \mathcal{G} \rangle$ stays invariant during the algorithm. Once the algorithm terminates, the set $\mathcal{G}$ satisfies the finite Buchberger criterion and hence is a Gröbner basis, due to Lemma 50. ∎

Example 54. Let $\mathcal{A} = \{a_0 < a_1 < ...\}$. Recall the ideal $\mathcal{I} \subseteq \mathbb{Q}[\mathcal{A}]$ from Example 24, that contains those polynomials without constant term, where the sum of coefficients of monomials of even degree is equal to the sum of coefficients of monomials of odd degree. As we mention in Remark 45, the base

$$\mathcal{B} = \left\{a_0^2 + a_0, a_0 - a_1\right\} \tag{6.4}$$

of $\mathcal{I}$, constructed in the proof of Theorem 13, is a Gröbner basis. Intuitively, the first polynomial in (6.4) is sufficient to reduce all univariate polynomials from $\mathcal{I}$, while the second one is sufficient to reduce all non-univariate polynomials to univariate ones.

For illustration of the operation of Algorithm 1, consider some other base $\mathcal{B}' = \left\{a_0 + a_1^2\right\}$ of $\mathcal{I}$. We explain how Algorithm 1, given $\mathcal{B}'$ as its input, can add to $\mathcal{B}'$ both polynomials from $\mathcal{B}$, thus computing a Gröbner basis. As the first step, the S-polynomial (equal to the second polynomial in $\mathcal{B}$)

$$a_0 + a_2^2 - (a_1 + a_2^2) = a_0 - a_1$$

is added to $\mathcal{B}'$, since it is reduced. Notably, an extra variable, for instance a_2, must be used to obtain this S-polynomial. In particular, we learn that $\mathcal{B}'$ is not a

Gröbner base, as the newly added polynomial $a_0 - a_1$ is reduced with respect to $\mathcal{B}'$. Then, in the second step, the S-polynomial

$$a_0 + a_1^2 + a_1(a_0 - a_1) = a_0 + a_0 a_1$$

reduces to the other polynomial $a_0 + a_0^2$ in $\mathcal{B}$, with respect to $\mathcal{B}' \cup \{a_0 - a_1\}$. Indeed, as the remainder of division of this S-polynomial by $a_0 - a_1$ we get the polynomial

$$a_0 + a_0 a_1 + a_0(a_0 - a_1) = a_0 + a_0^2,$$

which is reduced and hence is added to $\mathcal{B}'$. ◁

Remark 55. In this section we strengthen the total order assumption, and assume $\mathcal{A}$ to be well ordered, so that lexicographic order is well founded. To this effect, in Section 3 we replaced leading monomials with characteristic pairs. In this section this is not possible, as the proof of Lemma 49 does not adapt.

7. Decidability of ideal membership

As an immediate consequence of Section 6 we get:

Theorem 56. Ideal-memb in $\mathbb{K}[\mathcal{A}]$ *is decidable, for every computable field* $\mathbb{K}$ *and* ω*-well structured, well ordered and computable domain of variables* $\mathcal{A}$*.*

Proof. Given an instance $f \in \mathbb{K}[\mathcal{A}]$ and $\mathcal{B} \subseteq_{\text{fin}} \mathbb{K}[\mathcal{A}]$ we compute, using Algorithm 1, a Gröbner basis $\mathcal{G}$ of $\langle \mathcal{B} \rangle$. Then, relying on Lemmas 51 and 43, we compute some reduced polynomial f' obtainable by a sequence of division steps $f \longrightarrow_{\mathcal{G}}^{*} f'$. Relying on Lemma 44, we answer positively if $f' = \mathbf{0}$. ■

In the rest of this section we develop a method of transfering decidability of the ideal membership problem to other structures $\mathcal{A}$. The method is applied to the rational order $\mathcal{Q} = (Q, \leq_{\mathcal{Q}})$, which is ω-well structured and computable, but not well ordered.

Reduction game. Given a relational structure $\mathcal{A} = (A, \ldots)$ and $S, T \in \text{Age}(\mathcal{A})$ (source and target), by an (S,T)-*local embedding* we mean any partial bijection $S \to T$ (i.e., a bijection from a subset of S to a subset of T) that extends to an embedding in $\text{Emb}_{\mathcal{A}}$. The set of all (S,T)-local embeddings we call (S,T)-*profile* of $\mathcal{A}$.

We define a game $\mathcal{G}(\mathcal{A}, \mathcal{A}')$ induced by two relational structures $\mathcal{A} = (A, \ldots)$ and $\mathcal{A}' = (A', \ldots)$. The game is played by two players, Spoiler and Reducer, and consists of two rounds.

In the first round, Spoiler chooses $S, T \in \text{Age}(\mathcal{A})$, and Reducer responds by $S', T' \in \text{Age}(\mathcal{A}')$. We call S, S' *sources*, and T, T' *targets*. Source and target may intersect. Reducer wins the first round if there are bijections $\sigma : S' \to S$, $\tau : T \to T'$ such that (S', T')-profile of $\mathcal{A}'$ is equal to (S,T)-profile of $\mathcal{A}$, modulo σ and τ:

$$(S', T')\text{-prof}(\mathcal{A}') \;=\; \{\tau \circ \pi \circ \sigma \,|\, \pi \in (S,T)\text{-prof}(\mathcal{A})\}\,. \tag{7.1}$$

Otherwise Spoiler wins and the play ends.

In the second round, Spoiler chooses one of the structures $\mathcal{A}$, $\mathcal{A}'$, and a finite superset of target in this structure. Reducer responds similarly in the other structure. This step thus yields two *extended targets* $\overline{T} \in \text{Age}(\mathcal{A})$ and $\overline{T'} \in \text{Age}(\mathcal{A}')$ such that $T \subseteq \overline{T}$ and $T' \subseteq \overline{T'}$. Reducer wins the second round (and thus the whole play) if τ extends to a bijection $\overline{\tau} : \overline{T} \to \overline{T'}$ such that:

$$(S', \overline{T'})\text{-prof}(\mathcal{A}') \;=\; \left\{\overline{\tau} \circ \pi \circ \sigma \,\middle|\, \pi \in (S, \overline{T})\text{-prof}(\mathcal{A})\right\}. \tag{7.2}$$

We write $\mathcal{A} \ll \mathcal{A}'$ if Reducer has a computable winning strategy in $\mathcal{G}(\mathcal{A}, \mathcal{A}')$.

Lemma 57. $\mathcal{Q} \ll \omega^2$.

Proof. A subset $X \subseteq \omega^2$ we call *k-separated*, for $k \in \mathbb{N}$, if each two consecutive elements of X are separated by at least k elements of ω^2, and likewise there are at least k elements smaller than $\min(X)$.

Let $S, T \in \text{Age}(\mathcal{Q})$ be Spoiler's choice in the first round. Then Reducer's response is the following:

- as source S', choose the initial fragment of ω, of the same size as S (therefore $S' \subseteq \omega$);
- as target T', choose an arbitrary subset of non-zero multiplicities of ω, namely $T' \subseteq \{\omega \cdot b \,|\, b \in \omega, b > 0\}$, of the same size as T.

The unique monotonic bijections $\sigma : S' \to S$ and $\tau : T \to T'$ satisfy (7.1), namely (a proof below):

$$(S',T')\text{-prof}(\omega^2) \;=\; \{\tau \circ \pi \circ \sigma \,|\, \pi \in (S,T)\text{-prof}(\mathcal{Q})\}\,. \tag{7.3}$$

In the second round, if Spoiler chooses the structure ω^2 and plays extended target $\overline{T'} \subseteq \omega^2$, Reducer responds by arbitrary $\overline{T} \in \text{Age}(\mathcal{Q})$ so that τ extends to a monotonic bijection $\overline{\tau} : \overline{T} \to \overline{T'}$. This is always possible by denseness of $\leq_{\mathcal{Q}}$. Symmetrically, if Spoiler chooses the structure $\mathcal{Q}$ and plays $\overline{T} \in \text{Age}(\mathcal{Q})$, Reducer responds by arbitrary $\overline{T'} \subseteq \mathbb{N}^2$ so that τ again extends to a monotonic bijection $\overline{\tau} : \overline{T} \to \overline{T'}$. This is always possible as T' is k-separated for every $k \in \mathbb{N}$. Furthermore, Reducer chooses $\overline{T'}$ to be $|S'|$-separated. In both cases we get (7.2), namely

$$(S',\overline{T'})\text{-prof}(\omega^2) \;=\; \left\{\overline{\tau} \circ \pi \circ \sigma \,\middle|\, \pi \in (S,\overline{T})\text{-prof}(\mathcal{Q})\right\}. \tag{7.4}$$

Conditions (7.3) and (7.4) hold indeed, since τ is a monotonic bijection, and due to the following two observations:

Claim 58. *Every monotonic partial bijection $S \to \overline{T}$ extends to an embedding in* $\text{Emb}_{\mathcal{Q}}$.

Claim 59. *Every monotonic partial bijection $S' \to \overline{T'}$ extends to an embedding in* Emb_{ω^2}.

The former claim follows by homogeneity of $\mathcal{Q}$. For the latter claim, we first extend the mapping to S', which is possible as $\overline{T'}$ is $|S'|$-separated, and then define an extension on remaining arguments $\alpha \in \mathbb{N}^2 \setminus S'$ by $\alpha \mapsto \alpha + \beta$, for a sufficiently large ordinal β. ■

Corollary 61 below may be shown in the same way as Lemma 57, but it also follows from the general fact that the product of domains preserves the winner in the reduction game:

Lemma 60. *If $\mathcal{A}_1 \ll \mathcal{A}'_1$ and $\mathcal{A}_2 \ll \mathcal{A}'_2$ then $\mathcal{A}_1 \otimes \mathcal{A}_2 \ll \mathcal{A}'_1 \otimes \mathcal{A}'_2$.*

Proof. Assuming Reducer's strategies in games $\mathcal{G}(\mathcal{A}_1, \mathcal{A}'_1)$ and $\mathcal{G}(\mathcal{A}_2, \mathcal{A}'_2)$, we need to provide his strategy in $\mathcal{G}(\mathcal{A}_1 \otimes \mathcal{A}_2, \mathcal{A}'_1 \otimes \mathcal{A}'_2)$. Suppose Spoiler plays in the first round a source and target

$$S, T \in \text{Age}(\mathcal{A}_1 \otimes \mathcal{A}_2).$$

Let $S_1, T_1 \in \text{Age}(\mathcal{A}_1)$ and $S_2, T_2 \in \text{Age}(\mathcal{A}_2)$ be the projections. From Reducer's responses in $\mathcal{G}(\mathcal{A}_1, \mathcal{A}'_1)$ and $\mathcal{G}(\mathcal{A}_2, \mathcal{A}'_2)$, we get $S'_1, T'_1 \in \text{Age}(\mathcal{A}'_1)$, $S'_2, T'_2 \in \text{Age}(\mathcal{A}'_2)$ and bijections

$$\sigma_1 : S'_1 \to S_1 \qquad \tau_1 : T_1 \to T'_1$$
$$\sigma_2 : S'_2 \to S_2 \qquad \tau_2 : T_2 \to T'_2.$$

The products of these bijections,

$$(\sigma_1, \sigma_2) : S_1' \times S_2' \to S_1 \times S_2 \qquad (\tau_1, \tau_2) : T_1 \times T_2 \to T_1' \times T_2',$$

are used in REDUCER's response $S', T' \in \text{AGE}(\mathcal{A}_1' \otimes \mathcal{A}_2')$:

$$S' \;:=\; (\sigma_1, \sigma_2)^{-1}(S) \qquad T' \;:=\; (\tau_1, \tau_2)(T).$$

The restrictions of (σ_1, σ_2) and (τ_1, τ_2) to S' and T, respectively,

$$\sigma = (\sigma_1, \sigma_2)|S' \qquad \tau = (\tau_1, \tau_2)|T,$$

satisfy (7.1). Indeed, condition (7.1) follows directly by Lemma 8, and by the same condition for games $\mathcal{G}(\mathcal{A}_1, \mathcal{A}_1')$ and $\mathcal{G}(\mathcal{A}_2, \mathcal{A}_2')$.

Similarly we define REDUCER's strategy in the second round of $\mathcal{G}(\mathcal{A}_1 \otimes \mathcal{A}_2, \mathcal{A}_1' \otimes \mathcal{A}_2')$. ■

Corollary 61. $\mathcal{Q} \otimes \langle d \rangle \ll \omega^2 \otimes \langle d \rangle$.

Theorem 62. *Let $\mathbb{K}$ be a computable field and $\mathcal{A}, \mathcal{A}'$ two relational structures. If $\mathcal{A} \ll \mathcal{A}'$ then* IDEAL-MEMB IN $\mathbb{K}[\mathcal{A}]$ *reduces to* IDEAL-MEMB IN $\mathbb{K}[\mathcal{A}']$.

Proof. Given an instance of IDEAL-MEMB IN $\mathbb{K}[\mathcal{A}]$, i.e., a polynomial f and a finite basis $\mathcal{B}$, we compute an instance f', $\mathcal{B}'$ of IDEAL-MEMB IN $\mathbb{K}[\mathcal{A}']$ relying on REDUCER's strategy in the first round of $\mathcal{G}(\mathcal{A}, \mathcal{A}')$. We let SPOILER play $S := \text{VARS}(\mathcal{B})$, the set of variables appearing in $\mathcal{B}$, and $T := \text{VARS}(f)$. From the REDUCER's answer we get two subsets $S', T' \subseteq A'$ and bijections $\sigma : S' \to S$ and $\tau : T \to T'$. We use this bijections to rename variables in f and $\mathcal{B}$, thus obtaining f' and $\mathcal{B}'$.

Correctness of the reduction is shown using REDUCER's strategy in the second round. In one direction, suppose f belongs to the ideal generated by $\mathcal{B}$ in $\mathbb{K}[\mathcal{A}]$, i.e.,

$$f \;=\; h_1 \cdot \iota_1(g_1) \;+\; \ldots \;+\; h_m \cdot \iota_m(g_m), \tag{7.5}$$

for polynomials $g_1, \ldots, g_m \in \mathcal{B}$, polynomials $h_1, \ldots, h_m \in \mathbb{K}[\mathcal{A}]$ and embeddings $\iota_1, \ldots, \iota_m \in E$. We let SPOILER play $\overline{T}$ containing all variables appearing on the right in equality (7.5). Hence $T \subseteq \overline{T}$. From REDUCER's response we get $\overline{T'}$ and a bijection $\overline{\tau} : \overline{T} \to \overline{T'}$ that extends τ. We use σ and $\overline{\tau}$ to rename all variables in polynomials $h_1, \ldots, h_m$ and $g_1, \ldots, g_m$ appearing in the decomposition (7.5), which results in polynomials $h_1', \ldots, h_m' \in \mathbb{K}[\mathcal{A}']$ and $g_1', \ldots, g_m' \in \mathcal{B}'$. Furthermore, for every $i \in \{1 \ldots m\}$, we apply the equalities (7.1), (7.2) to the restriction $\pi_i = \iota_i|S$ to deduce that $\overline{\tau} \circ \pi_i \circ \sigma$ extends to some embedding $\iota_i' \in E'$. Therefore, we get:

$$f' \;=\; h_1' \cdot \iota_1'(g_1') \;+\; \ldots \;+\; h_m' \cdot \iota_m'(g_m'). \tag{7.6}$$

The reverse direction is shown similarly: assuming a decomposition (7.6), we let SPOILER choose $\mathcal{A}'$ in the second round, and use the bijections σ and τ obtained from REDUCER's response to get a decomposition (7.5). ■

Using Lemmas 9, 40, and 60, Theorems 56, 62, and Corollary 61 we get reduction of IDEAL-MEMB IN $\mathbb{K}[\mathcal{Q} \otimes \langle d \rangle]$ to IDEAL-MEMB IN $\mathbb{K}[\omega^2 \otimes \langle d \rangle]$. We easily generalize this reduction by invoking Lemma 60 several times. For conciseness, we write $\mathcal{A}^{(\ell)}$ for the ℓ-fold product: $\mathcal{A}^{(\ell)} := \underbrace{\mathcal{A} \otimes \ldots \otimes \mathcal{A}}_{\ell}$.

Corollary 63. *For every $d, \ell \geq 1$,* IDEAL-MEMB IN $\mathbb{K}[\mathcal{Q}^{(\ell)} \otimes \langle d \rangle]$ *reduces to* IDEAL-MEMB IN $\mathbb{K}[(\omega^2)^{(\ell)} \otimes \langle d \rangle]$ *and is therefore decidable.*

8. Application: reversible Petri nets with data

In this section we show that Theorem 56 is useful for showing decidability of the reachability problem for *reversible* Petri nets whose tokens carry data from a given domain (Theorem 66).

Petri nets with data. Among many presentations of Petri nets, we choose multiset rewriting, except that in place of finite multisets over $\mathcal{A}$ we conveniently take monomials $\textsc{Mon}[\mathcal{A}]$.

Let $\mathcal{A} = (A, \ldots)$ be a fixed relational structure. Any set of rewriting rules of the form

$$\mathcal{T} \subseteq \textsc{Mon}[\mathcal{A}] \times \textsc{Mon}[\mathcal{A}]$$

we call *monomial rewriting over* $\mathcal{A}$. Rewriting rules extend to *transitions* $\longrightarrow_{\mathcal{T}} \subseteq \textsc{Mon}[\mathcal{A}] \times \textsc{Mon}[\mathcal{A}]$ as follows: for every $g \in \textsc{Mon}[\mathcal{A}]$ and $(h, h') \in \mathcal{T}$, we have a transition

$$g \cdot h \longrightarrow_{\mathcal{T}} g \cdot h'.$$

The *reachability* relation is the transitive closure $\longrightarrow_{\mathcal{T}}^*$. In the sequel we always assume that $\mathcal{T}$ is orbit-finite, namely

$$\mathcal{T} \;=\; \textsc{Emb}_{\mathcal{A}}(T) \;\stackrel{\text{def}}{=}\; \{(\iota(h), \iota(h')) \,|\, \iota \in \textsc{Emb}_{\mathcal{A}},\ (h, h') \in T\}$$

for some finite set $T \subseteq_{\text{fin}} \textsc{Mon}[\mathcal{A}] \times \textsc{Mon}[\mathcal{A}]$. The set T serves as a *presentation* of $\mathcal{T}$ when it is input to algorithms. A *Petri net with data* $\mathcal{A}$ is just a monomial rewriting over $\mathcal{A} \otimes \langle d \rangle$ for some $d \in \mathbb{N}$:

Definition 64. *A Petri net with data $\mathcal{A}$, with d places, is an orbit-finite monomial rewriting $\mathcal{T}$ over $\mathcal{A} \otimes \langle d \rangle$.*

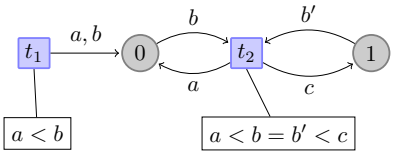
Example 65. In particular, monomial rewriting over $\langle d \rangle$ defines a plain (data-less) Petri net with d places $\{0 \ldots d-1\}$. The only embedding is identity, and hence $\mathcal{T}$ is finite. Monomial rewriting over $\mathcal{Q} \otimes \langle d \rangle$, where $\mathcal{Q} = (Q, \leq_{\mathcal{Q}})$ is the rational order, defines a Petri net with ordered data. According to Lemma 8, embeddings of $\mathcal{Q} \otimes \langle d \rangle$ are induced by some embedding ι of $\mathcal{Q}$ (i.e., a strictly increasing function $Q \to Q$), and preserve the second component, i.e., preserve places (for conciseness, instead of (a, i) we write ai):

$$ai \mapsto \iota(a)i.$$

Classical graphical presentation of Petri nets is extendable to Petri nets with data. For instance, consider a Petri net $\mathcal{T} = \textsc{Emb}_{\mathcal{Q}}(T)$ with $d = 2$ places, and with 2 transition rules $T = \{t_1, t_2\}$, where

$$t_1 = (\mathbf{1},\ a0 \cdot b0) \qquad t_2 = (b0 \cdot b1,\ a0 \cdot c1)$$

for some $a < b < c \in Q$ ($\mathbf{1}$ is the unit monomial). Petri net $\mathcal{T}$ is presented symbolically, using *variables* a, b, b', c ranging over Q:



Transition rule t_1 outputs two tokens with arbitrary but distinct rational values onto place 0. Transition t_2 inputs two tokens with the same value, say b, one from 0 and one from 1, and outputs two tokens: one token $a < b$ onto 0 and the other $c > b$ onto 1. ◁

$\mathcal{T}$ is *reversible* if it is closed under inverse: $c \longrightarrow_{\mathcal{T}}^* c'$ implies $c' \longrightarrow_{\mathcal{T}}^* c$ for every $c, c' \in \textsc{Mon}[\mathcal{A}]$. In such case we can assume, w.l.o.g., that T is closed under inverse: $(h, h') \in T \implies (h', h) \in T$.

Reachability problem. The *reachability problem* asks, given a Petri net $\mathcal{T}$ and two monomials, source and target $s, t \in \textsc{Mon}[\mathcal{A} \otimes \langle d \rangle]$, whether $s \longrightarrow_{\mathcal{T}}^{*} t$.

Theorem 66. *For every $d \geq 1$, the reachability problem in reversible Petri nets with d places, with data $\mathcal{A}$, reduces to* Ideal-memb in $\mathbb{Q}[\mathcal{A} \otimes \langle d \rangle]$ *(and also to* Ideal-memb in $\mathbb{Z}[\mathcal{A} \otimes \langle d \rangle]$*).*

Proof. The fundamental idea of the proof comes from [24]. Let $\mathcal{T} = \textsc{Emb}_{\mathcal{A}}(T)$ be a Petri net with data $\mathcal{A}$, with d places, where $T = \{(c_1, c_1'), \ldots, (c_n, c_n')\}$. Let $s, t \in \textsc{Mon}[\mathcal{A} \otimes \langle d \rangle]$ be source and target monomial. The reduction transformes s, t and T to the binomial $(t - s) \in \mathbb{Z}[\mathcal{A} \otimes \langle d \rangle]$ and the finite set of binomials $\mathcal{B} = \{(c_1' - c_1), \ldots, (c_n' - c_n)\} \subseteq \mathbb{Z}[\mathcal{A} \otimes \langle d \rangle]$. Correctness follows due to equivalence of the following three conditions:

(1) $c \longrightarrow_{\mathcal{T}}^{*} c'$;
(2) $(c' - c)$ belongs to the equivariant ideal generated by $\mathcal{B}$ in $\mathbb{Z}[\mathcal{A} \otimes \langle d \rangle]$;
(3) $(c' - c)$ belongs to the equivariant ideal generated by $\mathcal{B}$ in $\mathbb{Q}[\mathcal{A} \otimes \langle d \rangle]$;

The implications (1) $\Longrightarrow$ (2) and (3) $\Longrightarrow$ (1) are shown by reusing the proofs of Lemmas 1 and 2 in [24]. The remaining implication (2) $\Longrightarrow$ (3) is trivial. ■

Reduction of Theorem 66 is uniform in the number d of places. So is also the construction of Gröbner basis underlying the proof of Theorem 56, when instantiated to domains of the form $\mathcal{A} \otimes \langle d \rangle$. We thus deduce decidability of the reachability problem for reversible Petri nets for a wide range of domains $\mathcal{A}$, including $\mathcal{Q}$, equality data (a sub-case of $\mathcal{Q}$), and lexicographic products thereof (on the basis of Corollary 63):

Theorem 67. *For every $\ell \geq 1$, the reachability problem is decidable in reversible Petri nets with ℓ-fold nested ordered data $\mathcal{Q}^{(\ell)}$.*

Remark 68. Without reversiblity restriction, the reachability problem for Petri nets with ordered data is undecidable, while for Petri nets with equality data its status is still unknown.

9. Application: Orbit-finite systems of linear equations

We demonstrate that Theorem 56 is applicable to solving orbit-finite systems of linear equations, thus providing a simple proof of a generalisation of the main result of [15]. In the following, $\mathbb{K}$ is a field and $\mathcal{A} = (A, \ldots)$ an ω-well structured relational structure.

Solving a system of linear equations amount to checking if a given target vector b is a linear combination of a given set M of vectors. We generalise this setting to orbit-finite sets, by assuming that $b \in \textsc{Lin}(A^d)$ and that M is an equivariant orbit-finite family of vectors $M \subseteq \textsc{Lin}(A^d)$. For convenience we assume that the family M is *indexed* by a set A^n for some $n \in \mathbb{N}$, namely

$$M = (M_t \in \textsc{Lin}(A^d))_{t \in A^n}$$

(this is w.l.o.g. for equality and rational order domain [15, Sect.6]). Such a family M is equivariant if for every $\iota \in \textsc{Emb}_{\mathcal{A}}$ and $t \in A^n$,

$$\iota(M_t) = M_{\iota(t)},$$

i.e., $M_$ commutes with the action of embeddings, as the function $A^n \to \textsc{Lin}(A^d)$. Alternatively, M is an infinite $A^d \times A^n$ matrix over $\mathbb{K}$, whose every column M_t, being a vector in $\textsc{Lin}(A^d)$, has finitely many non-zero entries. A linear combination of M is of the form

$$r_1 \cdot M_{t_1} + \ldots + r_k \cdot M_{t_k}, \tag{9.1}$$

where $k \in \mathbb{N}$, $t_1, \ldots, t_k \in A^n$, and $r_1, \ldots, r_k \in \mathbb{K}$. We investigate the following problem:

LIN-SOLV$(\mathbb{K}, \mathcal{A})$**::**
Input: : a vector $b \in \text{LIN}(A^d)$ and an equivariant family
: $M = (M_t \in \text{LIN}(A^d))_{t \in A^n}$ of vectors.
Question: : Is b a linear combination of M?

Example 69. Let $\mathbb{K} = \mathbb{R}$, $\mathcal{A} = (A, =)$ the equality domain, $d = 1$ and $n = 2$, and consider an equivariant family of vectors $M = (M_{(a,b)} \in \text{LIN}(A))_{(a,b) \in A^2}$ defined as the following formal sums:

$$M_{(a,b)} = \begin{cases} a + 2 \cdot b & \text{when } a \neq b \\ 0 & \text{when } a = b \ . \end{cases}$$

For every $a \in A$, the singleton formal sum $a \in \text{LIN}(A)$ is a linear combination of M:

$$a \ = \ \frac{1}{2} \cdot M_{(a,b)} \ + \ \frac{1}{2} \cdot M_{(a,c)} \ - \ \frac{1}{3} \cdot M_{(b,c)} \ - \ \frac{1}{3} \cdot M_{(c,b)},$$

for any distinct $b, c \in A$ different than a. ◁

Theorem 70. LIN-SOLV$(\mathbb{K}, \mathcal{A})$ *reduces to* IDEAL-MEMB IN $\mathbb{K}[\mathcal{A}]$*, for every computable field* $\mathbb{K}$ *and relational structure* $\mathcal{A}$*.*

Theorem 70 yields decidability of LIN-SOLV$(\mathbb{K}, \mathcal{A})$, thus generalising [15, Theorem 4.4] which only considers equality structure. On the other hand, [15] provides an upper complexity bound.

Proof of Theorem 70. The reduction uses the same encoding ENC(_) : LIN$(A^d) \to \mathbb{K}[\mathcal{A}]$ as in Section 4. Consider an instance of LIN-SOLV$(\mathbb{K}, \mathcal{A})$, namely $M = (M_t)_{t \in A^n}$ and b. Relying on Lemma 27 we assume that A^n is represented by a finite set $B \subseteq_{\text{fin}} A^n$ such that $A^n = \text{EMB}_{\mathcal{A}}(B) \stackrel{\text{def}}{=} \{\iota(t) \,|\, \iota \in \text{EMB}_{\mathcal{A}}, t \in B\}$.

We construct an instance of IDEAL-MEMB IN $\mathbb{K}[\mathcal{A}]$ by taking the polynomial $f = \text{ENC}(b) \in \mathbb{K}[\mathcal{A}]$, and the equivariant $\mathcal{I}$ ideal generated by the finite set

$$\{\text{ENC}(M_t) \,|\, t \in B\} \ \subseteq \ \mathbb{K}[\mathcal{A}].$$

As encoding commutes with the action of embeddings, $\mathcal{I}$ is generated classically by the orbit-finite set $\{\text{ENC}(M_t) \,|\, t \in A^n\}$. We need to prove:

Claim 71. *b is a linear combination of M if and only if $f \in \mathcal{I}$.*

The 'only if' direction is immediate: ENC(_), as a linear function, maps a linear combination of vectors to a linear combination of their encoding polynomials. For the converse implication, suppose $f = \text{ENC}(b)$ belongs to $\mathcal{I}$, i.e.,

$$\text{ENC}(b) \ = \ r_1 \cdot h_1 \cdot \iota_1(\text{ENC}(M_{t_1})) + \ldots + r_1 \cdot h_k \cdot \iota_k(\text{ENC}(M_{t_k}))$$

for some $r_1, \ldots, r_k \in \mathbb{K}$, $h_1, \ldots, h_k \in \text{MON}[\mathcal{A}]$, $\iota_1, \ldots, \iota_k \in \text{EMB}_{\mathcal{A}}$ and $t_1, \ldots, t_k \in B$. As all monomials appearing in polynomials $\text{ENC}(M_{t_i})$ and $\text{ENC}(b)$ have the same degree, similarly as in the proof of Theorem 28 we can assume, w.l.o.g., that all monomials h_i are unit monomials, i.e.,

$$\text{ENC}(b) \ = \ r_1 \cdot \iota_1(\text{ENC}(M_{t_1})) + \ldots + r_1 \cdot \iota_k(\text{ENC}(M_{t_k})).$$

As encoding commutes with the action of embeddings and with scalar multiplication, we get

$$\text{ENC}(b) \ = \ \text{ENC}(r_1 \cdot \iota_1(M_{t_1})) + \ldots + \text{ENC}(r_1 \cdot \iota_k(M_{t_k})),$$

and by injectivity of encoding and equivariance of M we obtain

$$b \ = \ r_1 \cdot M_{\iota_1(t_1)} + \ldots + r_1 \cdot M_{\iota_k(t_k)},$$

i.e. b is a linear combination of M, as required. ■

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